

# FreeMoCap

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## Abstract

## Introduction

Motion capture provides detailed quantitative measurements of human movement, but deploying it beyond specialized laboratories remains difficult. Technologies introduced into clinical practice or real-world environments must balance measurement detail with accessibility, ease of use, and cost (Kerr *et al.*, 2018; Buckley *et al.*, 2019). For example, while marker-based motion capture (often considered a gold-standard in measurement analysis) can support detailed biomechanical analyses in a laboratory setting (Delp *et al.*, 2007), its cost, hardware, and technical expertise requirements can limit translation into widespread use (Sinclair, Hebron and Taylor, 2014; Camomilla, Dumas and Cappozzo, 2017). A gap therefore remains between the detailed kinematics available in a motion analysis laboratory and the movement information that can be feasibly collected in clinical and real-world settings (Wei and Wang, 2024).

### Approaches to early and modern markerless motion capture

Markerless motion capture (i.e., the extraction of kinematic data from video without physical markers) developed in part as an attempt to close this gap. In early markerless approaches, the representation of the body was often closely tied to the method used to recover its pose from images. Pre-defined or subject-specific body models were matched to image-derived information such as edges, silhouettes, or reconstructed visual hulls, with body pose estimated from the fitted model (Gavrila and Davis, 1996; Rosenhahn *et al.*, 2006; Corazza *et al.*, 2006). Substituting any one of these components could therefore require substantial changes to the rest of the system.

The emergence of deep-learning-based pose estimation (e.g., DeepPose (Toshev and Szegedy, 2014) and OpenPose (Cao *et al.*, 2019)) marked a fundamental shift. Rather than fitting a body model to image features, pose estimation models could infer anatomical landmark locations directly from images. These methods have since become central to modern markerless motion capture (Nagorny *et al.*, 2025), where two-dimensional keypoints estimated independently from multiple camera views can be triangulated to reconstruct three-dimensional movement.

### Compositional pipelines expand methodological choice and add complexity to validation

Modern systems might now be better described as compositional pipelines, assembling different combinations of hardware, pose estimation models, reconstruction approaches, and downstream analysis methods. For example, Theia3D (Kingston, ON, Canada) uses six or more hardware-synchronized cameras together with proprietary pose estimation and reconstruction methods to collect 3D data (Kanko *et al.*, 2021), while OpenCap, an open-source alternative, uses synchronized iOS devices, supports OpenPose and HRNet for pose estimation, and applies a learned marker-augmentation model after triangulation to infer anatomical marker locations for biomechanical modeling (Uhlrich *et al.*, 2023). Reusable open-source platforms such as Pose2Sim (Pagnon, Domalain and Reveret, 2022) and PosePipe (Cotton, 2022), alongside numerous laboratory-specific implementations (Nakano *et al.*, 2020; Needham *et al.*, 2022; Yang and Jung, 2025), further illustrate the range of possible configurations.

This flexibility, however, adds complexity to validation. Implementation and run-time choices at each stage of the pipeline can affect the resulting measurements. We can examine pose estimation as a clear

illustration. Each pose estimation algorithm can differ in performance across participants, poses, movements, and recording conditions, reflecting their training data and anatomical representations (Needham *et al.*, 2021; Roggio *et al.*, 2024), and not all are equally suited to biomechanical applications (Seethapathi *et al.*, 2019). Similar dependencies can arise from choices in camera configuration, reconstruction, filtering, and downstream modeling. Validation of markerless motion capture therefore benefits from characterizing how performance varies across methodological choices and measurement contexts, rather than treating accuracy as a fixed property of the system. This is especially important for low-cost systems intended for broader deployment, where performance may differ across tasks, movement intensities, biomechanical outcomes, and the acquisition conditions that make such accessibility possible.

Accessible markerless motion capture therefore requires more than low-cost hardware and acceptable accuracy. The methods used to produce the measurements, together with the procedures used to validate them, should be sufficiently open and reproducible that performance can be characterized, compared, and re-evaluated as components, tasks, and use cases change.

### **Characterizing an accessible, modular system across tasks and pose estimation backends**

Here we validate and characterize FreeMoCap, a free and open-source markerless motion-capture framework developed to make 3D kinematic measurement available without specialized hardware or programming expertise. FreeMoCap reconstructs 3D kinematics from synchronized video acquired with consumer-grade cameras through a graphical user interface and runs locally without requiring a cloud service or internet connection. Its major components, including synchronized acquisition, camera calibration, 2D pose estimation, 3D reconstruction, and post-processing, are independently maintained and separable, allowing individual stages to be modified or exchanged.

Prior studies have validated FreeMoCap under specific configurations and movement tasks. Darici *et al.* reported strong agreement for spatiotemporal gait measures (Darici, Cabak and Wong, 2026), while Minogue *et al.* found strong sagittal-plane agreement during squatting and countermovement jumping, with larger errors for some joint angles (Minogue *et al.*, 2025). Núñez-Lisboa *et al.* further showed that FreeMoCap could detect experimentally induced changes in gait mechanics (Núñez-Lisboa *et al.*, 2024). Each of these studies used MediaPipe for pose estimation and examined dynamic, large-amplitude movement, leaving open how performance depends on the pose estimation backend and how it behaves when the movement of interest is small.

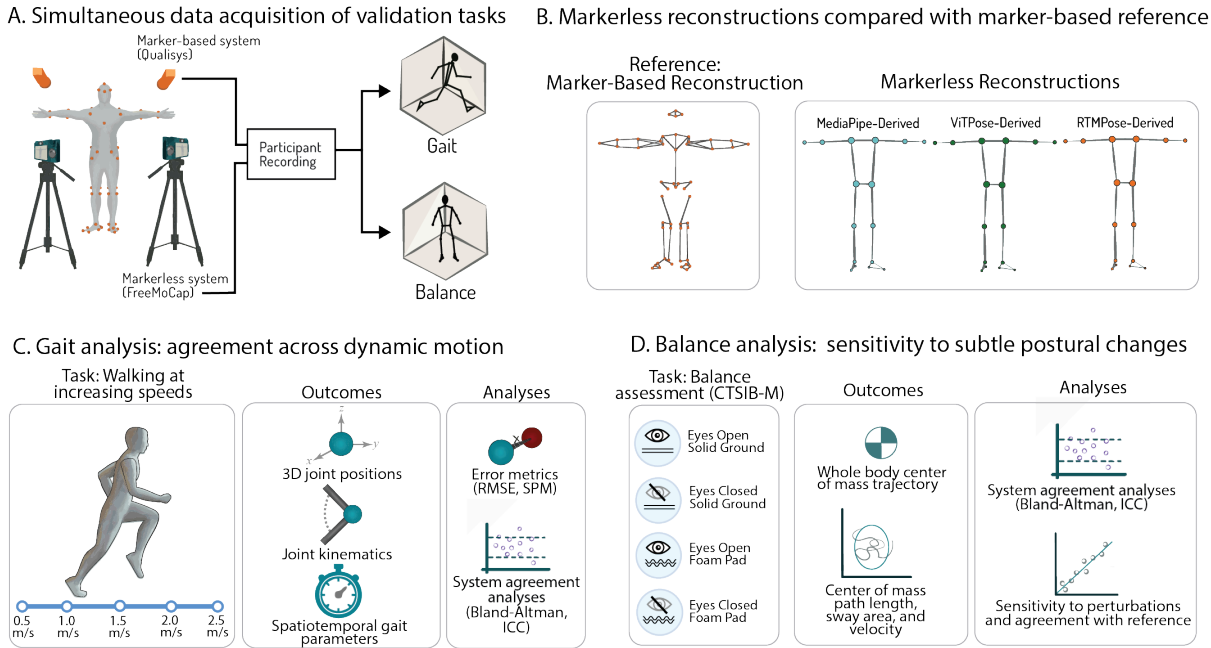
In the present study, we characterize a six-webcam FreeMoCap system, assembled for a total camera cost below \$150, against simultaneously recorded marker-based motion capture during two tasks: gait over a range of walking speeds and standing balance during the Modified Clinical Test of Sensory Interaction on Balance (CTSIB-M). These tasks impose substantially different measurement demands, from large, dynamic, cyclic movement to subtle postural adjustments. Each dataset was reconstructed using three pose-estimation backends (MediaPipe, RTMPose, and ViTPose), with acquisition, calibration, and reconstruction held constant. We evaluate how measurement validity varies across pose estimation backends, movement conditions, and biomechanical outcomes, and whether an inexpensive webcam-based system can provide meaningful gait and balance measurements.

## **Results**

### **Reconstruction and comparison of 3D motion capture data**

Participants completed both a gait and balance assessment while being simultaneously recorded by the markerless and marker-based motion capture systems to characterize the ability of the markerless system to capture both dynamic and subtle movements (Figure 1). Joint centers calculated from the marker-based data served as the reference for comparison. The markerless data were produced using a single FreeMoCap

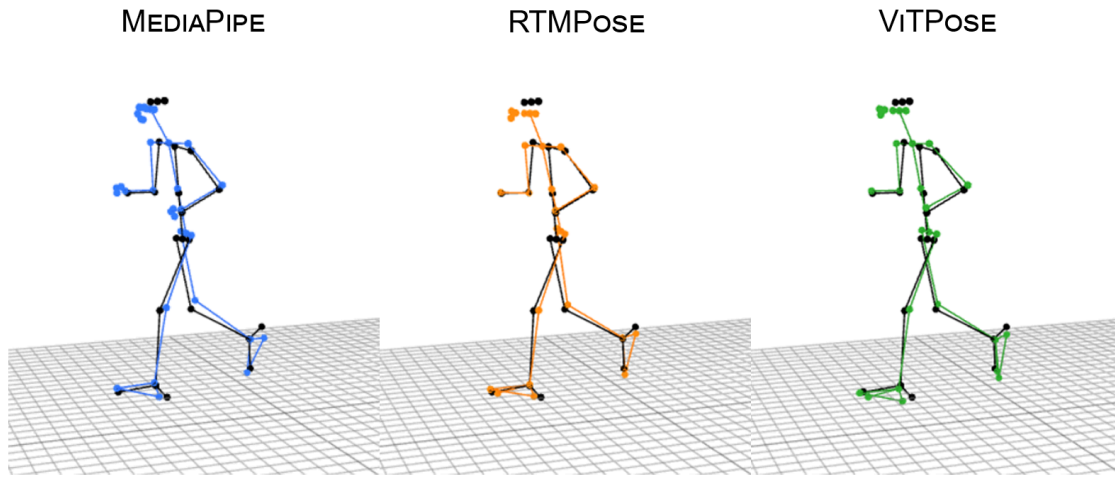
pipeline in which 2D pose estimation (the detection of body keypoints within each camera view) was treated as an interchangeable module. Each trial was processed separately using three pose estimation backends: MediaPipe (Lugaresi *et al.*, 2019), RTMPose (Jiang *et al.*, 2023), and ViTPose (Xu *et al.*, 2022). All other stages of the pipeline, including multi-camera synchronization, calibration, and triangulation, were held constant. Each trial therefore yielded three sets of 3D markerless data that differed only in the pose estimator used to generate the underlying 2D keypoints.



**Figure 1.** Overview of the study design and validation analyses. **A.** Markerless and marker-based motion capture data were acquired simultaneously using six consumer webcams and a reference marker-based system. Participants completed two movement assessments: treadmill walking across increasing speeds to evaluate dynamic gait performance and the Modified Clinical Test of Sensory Interaction on Balance (CTSIB-M) to evaluate subtle postural adjustments under progressively more challenging sensory conditions. **B.** Markerless data were reconstructed using three pose estimation backends: MediaPipe, ViTPose, and RTMPose, and compared with the marker-based reference. **C.** Gait validity was evaluated across increasing walking speeds using 3D joint trajectories, joint kinematics, and spatiotemporal gait parameters. Analyses included error metrics, statistical parametric mapping, Bland-Altman analysis, and intraclass correlation coefficients. **D.** Balance validity was evaluated during the CTSIB-M using whole-body center-of-mass trajectories, path length, 95% confidence ellipse area, and mean horizontal velocity. Analyses assessed both agreement with the marker-based reference and sensitivity to visual, proprioceptive, and combined sensory perturbations.

### Reconstructing dynamic motion with gait

Participants completed two treadmill walking trials in which walking speed was progressively increased (from 0.5 to 2.5 m/s at 0.5 m/s increments), while simultaneously recorded by the markerless and marker-based motion capture systems. Representative 3D reconstructions from each pose estimation backend are shown alongside joint centers calculated from the marker-based reference in Figure 2.



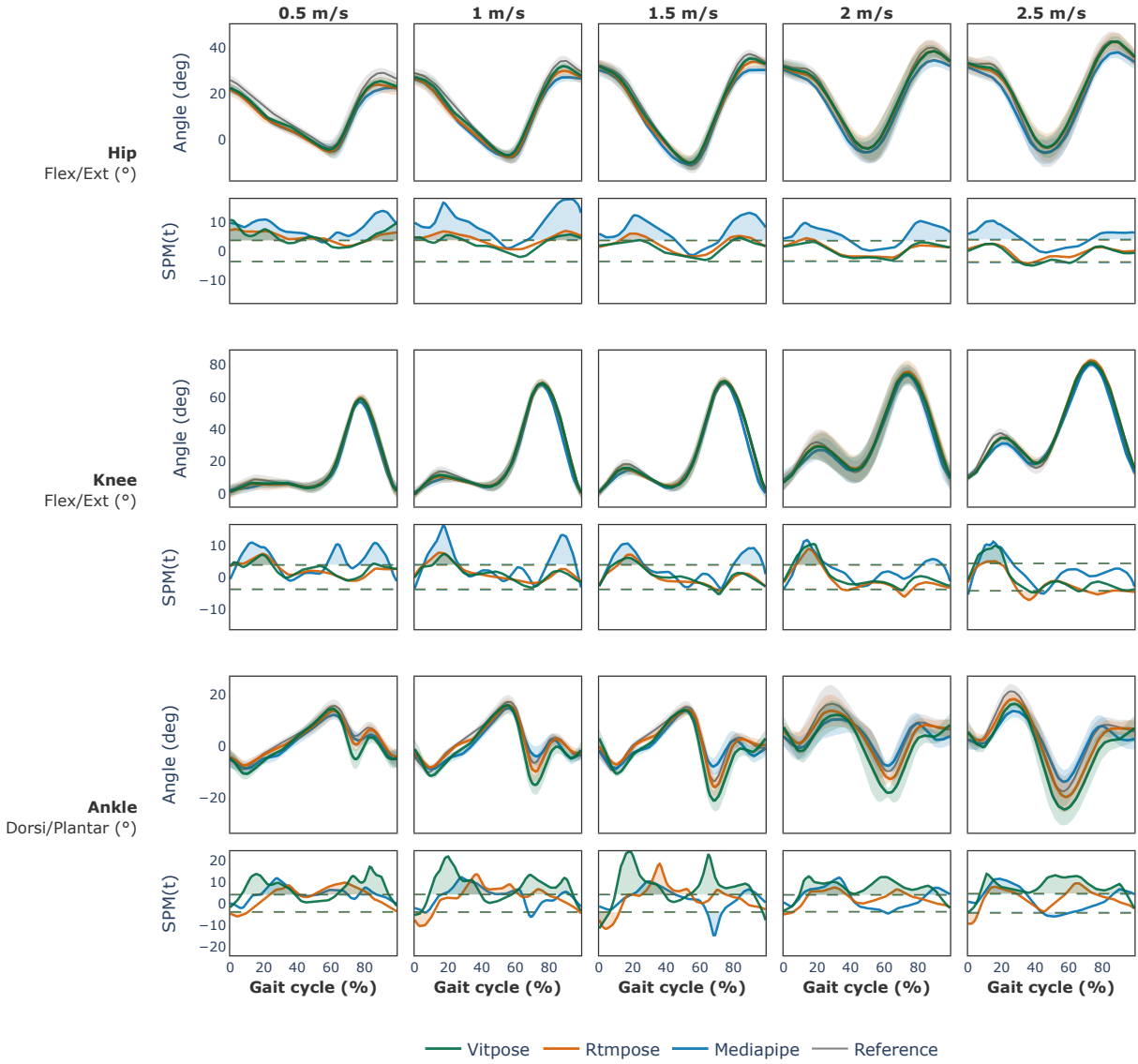
**Figure 2.** Example of markerless motion capture data using each pose estimation backend. **Left:** MediaPipe (blue); **Middle:** RTMPose (orange); **Right:** ViTPose (green). Joint centers calculated from the marker-based reference skeleton are displayed in black on each panel.

[PLACEHOLDER: Video of multiple skeletons walking, maybe with the joint angles and gait events overlaid]

#### **Gait: Joint kinematics error**

Sagittal-plane lower-body joint kinematics were calculated from gait cycle-normalized strides for each pose estimation backend and walking speed. Regions of significant difference between the markerless and marker-based joint angle trajectories were identified using statistical parametric mapping (SPM) two-tailed t-tests and are shown in Figure 3. We also quantified joint angle error across joints, pose estimation backends, and walking speed using RMSE.

SPM paired t-tests ( $\alpha = 0.05$ ) revealed common suprathreshold clusters during early stance at all three joints, with on backends. At the ankle, ViTPose-derived angles exhibited widespread differences spanning much of the gait cycle. At the hip and knee, MediaPipe-derived angles exhibited suprathreshold clusters whose magnitude and duration decreased as walking speed increased.



**Figure 3.** Comparison of sagittal hip, knee, and ankle joint angles across walking speeds (0.50-2.50 m/s). Angles derived from MediaPipe (blue), RTMPose (orange), and ViTPose (green) are shown alongside the marker-based reference (grey). Shaded error bars indicate  $\pm 1$  SD across strides and the gait cycle is normalized to 0-100%. Statistical parametric mapping (SPM) results are shown in the lower panels. Dashed lines indicate the critical threshold ( $t^*$ ), and shaded regions denote time intervals where differences from the marker-based reference are statistically significant ( $p < .05$ ).

Sagittal-plane joint angle RMSE remained below  $5^\circ$  for most conditions, with the primary exception of ViTPose-derived ankle angles at higher walking speeds (Table 1). These errors increased with speed at the knee and ankle, while hip angle error remained relatively stable. ViTPose produced the lowest hip and knee errors overall but showed a consistent plantarflexed offset at the ankle, whereas RTMPose produced the most accurate ankle angles.

**Table 1.** Joint angle RMSE (°) by speed. Values represent mean  $\pm$  SD across trials compared to Qualisys.

| Joint                   | Backend   | 0.5 m/s       | 1 m/s         | 1.5 m/s       | 2 m/s         | 2.5 m/s       |
|-------------------------|-----------|---------------|---------------|---------------|---------------|---------------|
| Hip Flex/Ext            | MediaPipe | 4.1 $\pm$ 1.2 | 4.1 $\pm$ 0.8 | 4.0 $\pm$ 1.0 | 4.0 $\pm$ 1.2 | 4.0 $\pm$ 1.2 |
|                         | RTMPose   | 3.9 $\pm$ 1.7 | 3.4 $\pm$ 1.8 | 3.1 $\pm$ 1.5 | 2.9 $\pm$ 1.5 | 3.2 $\pm$ 1.7 |
|                         | ViTPose   | 3.0 $\pm$ 0.7 | 2.4 $\pm$ 0.6 | 2.3 $\pm$ 0.7 | 2.7 $\pm$ 0.9 | 2.9 $\pm$ 1.0 |
| Knee Flex/Ext           | MediaPipe | 3.1 $\pm$ 0.7 | 3.5 $\pm$ 0.7 | 3.8 $\pm$ 1.1 | 3.8 $\pm$ 0.9 | 3.9 $\pm$ 0.8 |
|                         | RTMPose   | 2.2 $\pm$ 0.6 | 2.2 $\pm$ 0.6 | 2.4 $\pm$ 0.7 | 3.1 $\pm$ 0.8 | 3.6 $\pm$ 1.1 |
|                         | ViTPose   | 2.0 $\pm$ 0.3 | 1.8 $\pm$ 0.6 | 2.1 $\pm$ 0.9 | 2.7 $\pm$ 1.0 | 3.2 $\pm$ 1.2 |
| Ankle Dorsi/<br>Plantar | MediaPipe | 2.6 $\pm$ 0.9 | 3.1 $\pm$ 0.8 | 3.4 $\pm$ 0.8 | 3.9 $\pm$ 1.1 | 4.7 $\pm$ 0.8 |
|                         | RTMPose   | 1.6 $\pm$ 0.4 | 2.0 $\pm$ 0.4 | 2.1 $\pm$ 0.5 | 2.7 $\pm$ 0.9 | 3.2 $\pm$ 1.1 |
|                         | ViTPose   | 3.5 $\pm$ 0.9 | 4.3 $\pm$ 0.9 | 4.7 $\pm$ 0.9 | 5.8 $\pm$ 1.4 | 6.2 $\pm$ 1.4 |

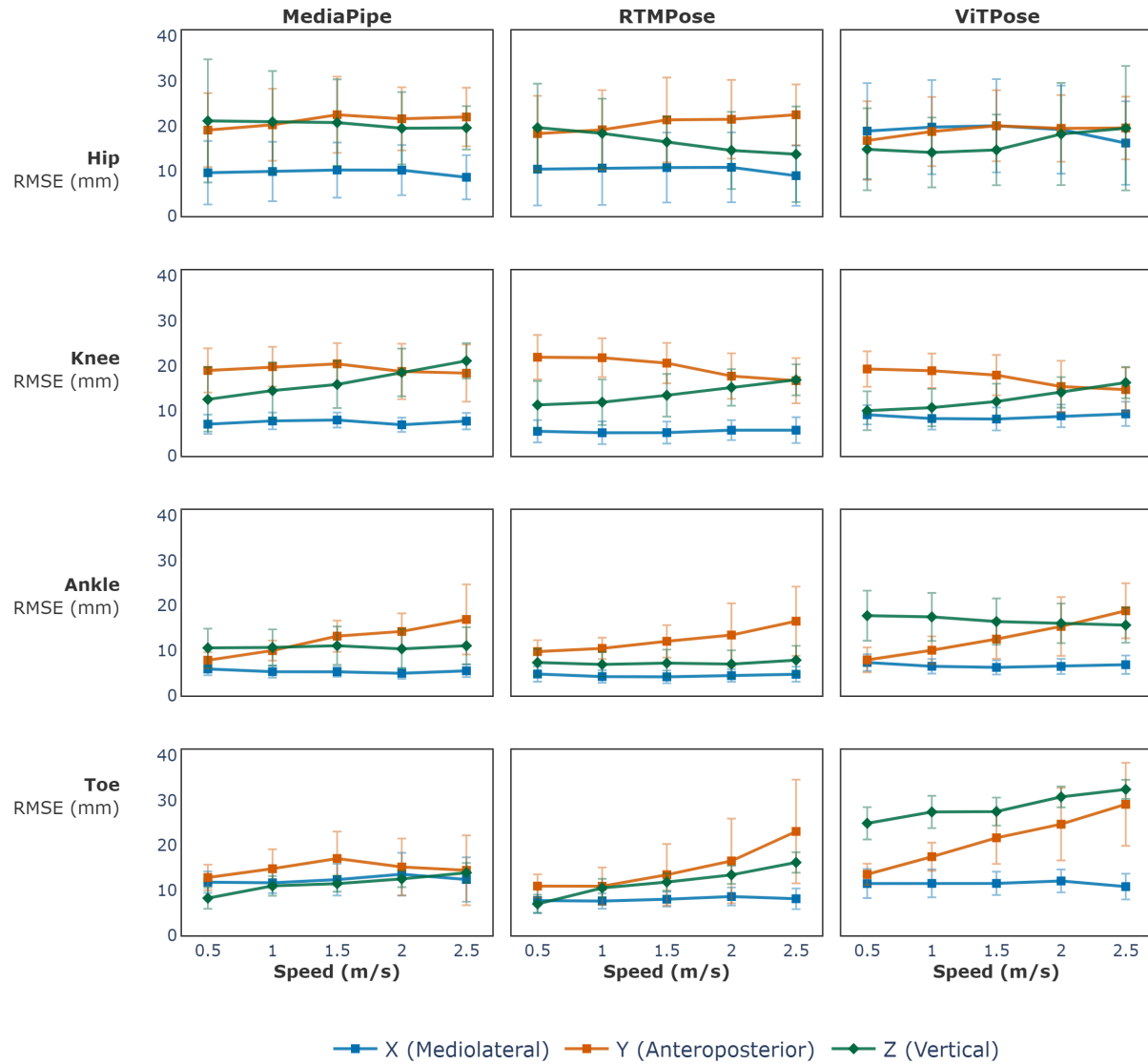
#### Gait: Joint position error

Reconstructed joint center position errors for each lower-limb joint, walking speed, axis, and pose estimation backend are summarized in Figure 4. Full joint center trajectories for every joint, backend, and walking speed are shown along the mediolateral (Figure 4—figure supplement 1), anteroposterior (Figure 4—figure supplement 2), and vertical (Figure 4—figure supplement 3) axes. Corresponding RMSE values are provided in Appendix 1—table 1, Appendix 1—table 2, and Appendix 1—table 3.

Joint-center RMSE was generally below 30 mm and was lowest in the mediolateral (ML) direction. Across joints, the hip exhibited the largest overall error, with an RMSE of approximately 20 mm in both the anteroposterior (AP) and vertical directions. However, hip error was largely unaffected by walking speed.

At more distal joints, error generally increased with walking speed, particularly in the AP and vertical directions, although this pattern varied across joints, axes, and pose estimation backends. For example, AP knee error decreased with speed for RTMPose and ViTPose-derived trajectories, while vertical ankle error remained relatively stable across speeds for all three backends.

Across pose estimation backends, RTMPose generally produced the lowest joint center trajectory error, whereas ViTPose produced the highest, particularly in the vertical direction. These differences were most pronounced at the ankle and toe, where ViTPose-derived trajectories showed a consistent vertical offset relative to the marker-based reference.



**Figure 4.** Trajectory RMSE (mm) across joints, axes, and walking speeds for each pose estimation backend. RMSE was calculated per gait-cycle-normalized stride and averaged across strides within each trial. Rows correspond to joint centers (hip, knee, ankle, toe) and columns to pose estimation backends (MediaPipe, RTMPose, ViTPose). Error is shown separately for the mediolateral (blue), anteroposterior (orange), and vertical (green) directions. Points represent mean RMSE across all trials; error bars indicate  $\pm 1$  SD.

#### Gait: Spatiotemporal parameter agreement

Heel-strike and toe-off events were identified and used to calculate stance duration, swing duration, stride duration, stride length, and step length. Agreement with the marker-based reference was assessed for each parameter using Bland-Altman bias and 95% limits of agreement (LoA), together with intraclass correlation coefficients (ICC). ICC values below 0.50 were interpreted as poor agreement, values from 0.50 to 0.75 as moderate, values from 0.75 to 0.90 as good, and values above 0.90 as excellent.

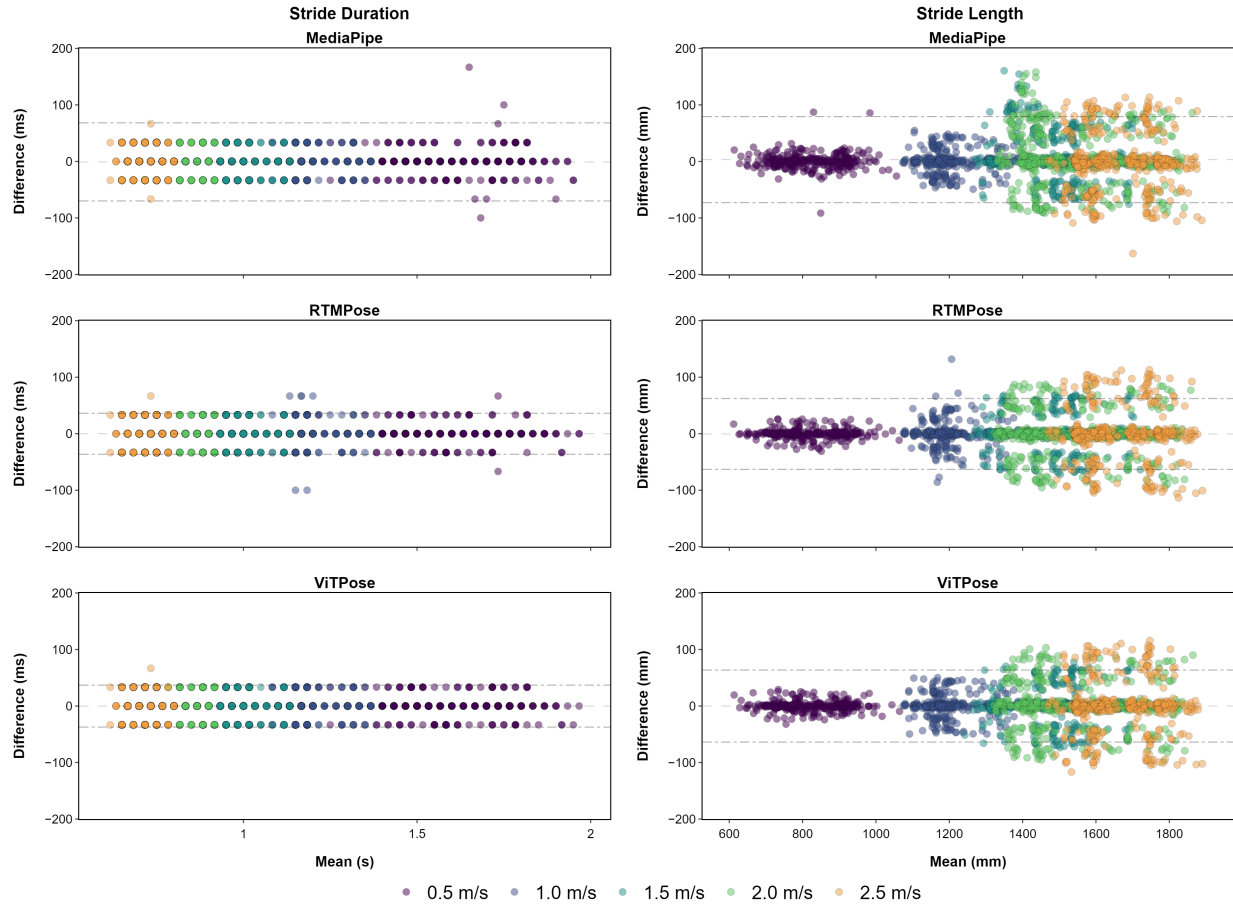
When pooled across walking speeds, spatiotemporal gait parameters showed minimal bias and excellent agreement with the marker-based reference (ICC > 0.90; Table 2). Across pose estimation backends,

ViTPose-derived data produced near-zero bias for all parameters. RTMPose and ViTPose-derived data both showed relatively narrow LoA, whereas MediaPipe-derived data showed larger deviations and wider LoA.

**Table 2.** Bland-Altman agreement statistics for spatiotemporal gait parameters across all walking speeds. Bias and limits of agreement (LoA) are reported in the native units of each metric. ICC = intraclass correlation coefficient (2,1) with 95% confidence interval.

| Metric               | Backend   | Bias  | Lower LoA | Upper LoA | ICC (95% CI)         |
|----------------------|-----------|-------|-----------|-----------|----------------------|
| Stride Length (mm)   | MediaPipe | -0.29 | -68.98    | +68.41    | 0.993 (0.990, 0.990) |
|                      | RTMPose   | -0.41 | -63.05    | +62.23    | 0.994 (0.990, 0.990) |
|                      | ViTPose   | -0.07 | -63.73    | +63.59    | 0.994 (0.990, 0.990) |
| Step Length (mm)     | MediaPipe | +0.70 | -68.64    | +70.04    | 0.972 (0.970, 0.970) |
|                      | RTMPose   | -0.31 | -62.18    | +61.56    | 0.977 (0.980, 0.980) |
|                      | ViTPose   | +0.16 | -66.01    | +66.33    | 0.974 (0.970, 0.980) |
| Stance Duration (ms) | MediaPipe | +4.78 | -57.64    | +67.20    | 0.991 (0.990, 0.990) |
|                      | RTMPose   | +2.48 | -34.09    | +39.06    | 0.997 (1.000, 1.000) |
|                      | ViTPose   | -0.93 | -37.76    | +35.89    | 0.997 (1.000, 1.000) |
| Swing Duration (ms)  | MediaPipe | -5.35 | -49.27    | +38.58    | 0.900 (0.880, 0.910) |
|                      | RTMPose   | -2.56 | -38.92    | +33.81    | 0.936 (0.930, 0.940) |
|                      | ViTPose   | +0.81 | -37.13    | +38.75    | 0.932 (0.930, 0.940) |
| Stride Duration (ms) | MediaPipe | -0.83 | -69.85    | +68.20    | 0.993 (0.990, 0.990) |
|                      | RTMPose   | -0.08 | -36.36    | +36.20    | 0.998 (1.000, 1.000) |
|                      | ViTPose   | -0.12 | -37.23    | +36.99    | 0.998 (1.000, 1.000) |

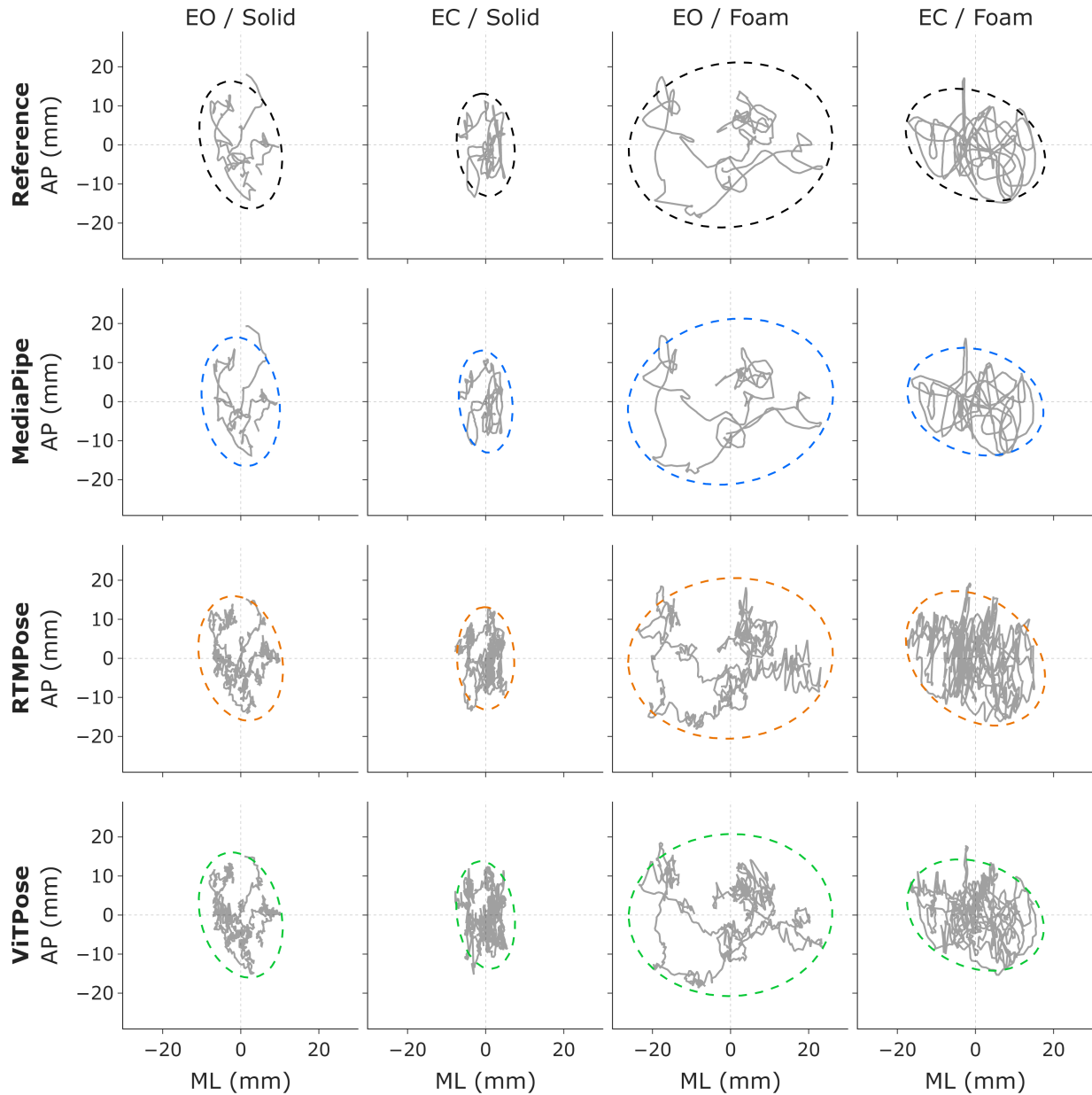
Agreement for spatial parameters was speed-dependent, with LoA widening markedly as walking speed increased (e.g., ViTPose step length:  $\pm 19$  to  $\pm 88$  mm). Step length ICC decreased from strong agreement at slow speeds to moderate agreement at the fastest speed, whereas stride length remained excellent across speeds. Temporal parameters showed a different pattern: bias and limits of agreement remained relatively consistent across speeds, with differences clustering at multiples of the 33 ms frame interval (Figure 5), but ICC still declined at higher speeds, most sharply for stance duration. Across spatial and temporal parameters, MediaPipe-derived data produced the widest LoA and lowest ICC values, consistent with the pooled results (Table 2). Full speed-stratified Bland-Altman and ICC results are provided for spatial parameters in Appendix 1—table 4 and temporal parameters in Appendix 1—table 5.



**Figure 5.** Bland-Altman plots of stride duration (left) and stride length (right) agreement between markerless and marker-based systems. Rows correspond to each pose estimation backend (MediaPipe, RTMPose, ViTPose), and colors indicate walking speed (0.50-2.50 m/s). Dashed lines represent bias, and dash-dot lines indicate 95% limits of agreement. Stride duration differences are reported in milliseconds and stride length differences in millimeters.

### Balance: Analyzing postural stability using center of mass

We next evaluated whether the same markerless reconstructions captured differences in postural stability during standing balance tasks. Participants completed two trials of the Modified Clinical Test of Sensory Interaction on Balance (CTSIB-M). During each trial, participants stood for 60 seconds under four conditions that varied visual input and support surface stability: 1) eyes open on a firm surface; 2) eyes closed on a firm surface; 3) eyes open on a foam surface; and 4) eyes closed on a foam surface. Center of mass (COM) trajectories were estimated from each pose estimation backend and from the marker-based reference. Postural stability was then quantified using COM path length, 95% confidence ellipse area, and mean three-dimensional COM velocity. Representative COM trajectories projected onto the ground plane for each CTSIB-M condition are shown in Figure 6.



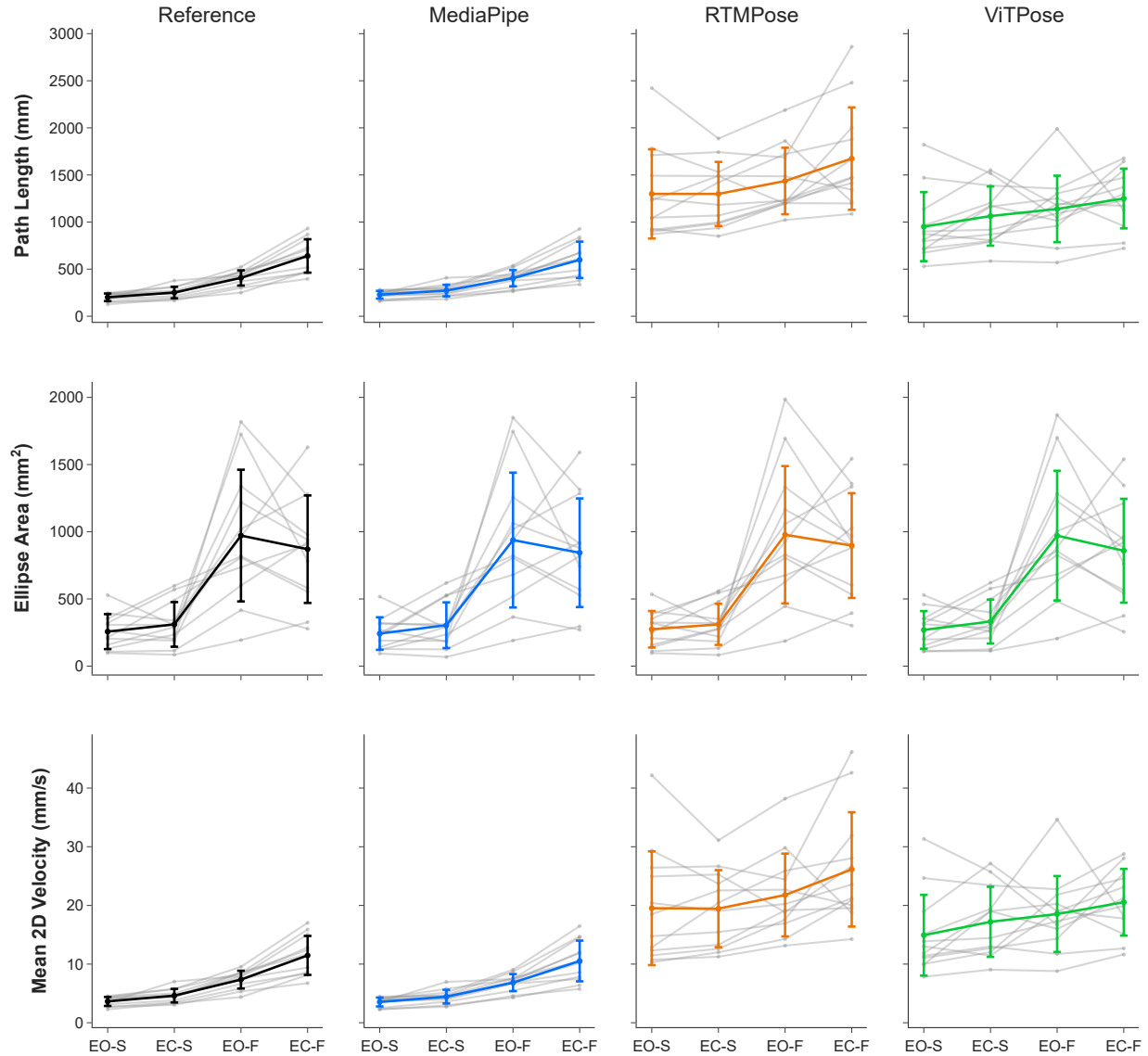
**Figure 6.** Center-of-mass trajectories in the horizontal plane for a representative trial across the four CTSIB-M conditions. Rows show trajectories derived from the marker-based reference and each pose estimation backend; columns show the eyes open/solid, eyes closed/solid, eyes open/foam, and eyes closed/foam conditions. Dashed lines indicate the 95% confidence ellipse for each trajectory. Although the overall spatial extent of sway was broadly similar across systems, RTMPose and ViTPose-derived trajectories exhibited greater frame-to-frame variability than the reference and MediaPipe-derived trajectories. AP = anteroposterior; ML = mediolateral; EO = eyes open; EC = eyes closed.

[PLACEHOLDER: Video of skeletons doing balance assessment - maybe 1 per tracker? hard to fit all 4 onto one video]

### Balance: Comparing postural stability

Mean posturographic metrics and individual trial values for each balance condition and 3D dataset are shown in Figure 7. Exact values are provided in Appendix 2—table 1.

Although 95% confidence ellipse area was comparable across pose-estimation backends and the marker-based reference, only MediaPipe-derived COM path length and mean horizontal velocity closely matched the reference and preserved the expected separation across progressively more challenging balance conditions. In contrast, RTMPose and ViTPose-derived path length and velocity overestimated the reference and did not clearly differentiate among conditions.

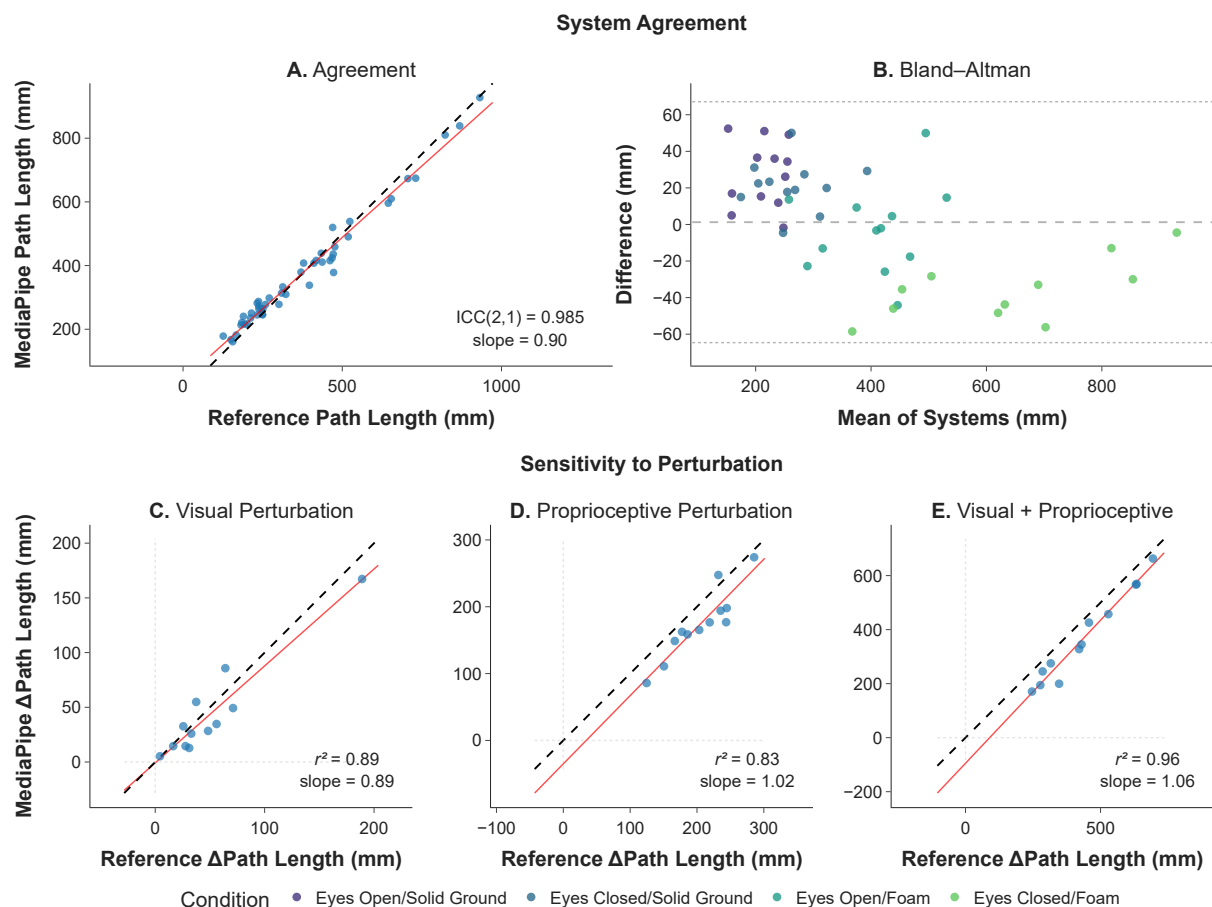


**Figure 7.** Center-of-mass balance metrics across CTSIB-M conditions for the marker-based reference and each pose-estimation backend. Rows show path length, 95% confidence ellipse area, and mean horizontal COM velocity; columns show the reference, MediaPipe, RTMPose, and ViTPose-derived results. Gray lines represent individual trials, and colored lines with error bars represent the group mean and variability. MediaPipe-derived path length and mean horizontal velocity closely followed the reference and preserved the progressive separation among balance conditions, whereas RTMPose and ViTPose produced elevated estimates with less distinct condition separation. Ellipse area was comparatively similar across systems.

[PLACEHOLDER: Potentially the video comparison of pose estimation software on quiet standing to show how shaky rtmppose/vitppose are]

## Balance: Agreement and sensitivity of MediaPipe-derived 3D data

We further examined the validity of MediaPipe-derived COM path length by evaluating two properties: agreement, defined as the correspondence between absolute path-length values from the two systems; and sensitivity, defined as the ability to reproduce the change in path length associated with three sensory perturbations. The sensory perturbations were comprised of: removal of visual information (Eyes Closed/Solid Ground path length minus Eyes Open/Solid Ground path length), altered proprioceptive information (Eyes Open/Foam path length minus Eyes Open/Solid Ground path length), and combined visual and proprioceptive perturbation (Eyes Closed/Foam path length minus Eyes Open/Solid Ground path length) (Figure 8).



**Figure 8.** Agreement and sensitivity of MediaPipe-derived center of mass path length relative to the marker-based reference. **A.** Absolute agreement between MediaPipe-derived and reference path length across all balance conditions; the dashed line indicates identity and the solid line shows the fitted regression. ICC and regression slope summarize absolute agreement. **B.** Bland-Altman analysis of absolute path length, with points colored by balance condition; the central dashed line indicates the mean difference between systems, and the outer dotted lines indicate the 95% limits of agreement. **C-E.** Sensitivity to path length changes from: removal of visual information (C: Eyes Closed/Solid Ground minus Eyes Open/Solid Ground), altered proprioceptive information (D: Eyes Open/Foam minus Eyes Open/Solid Ground), and combined visual and proprioceptive perturbation (E: Eyes Closed/Foam minus Eyes Open/Solid Ground). Positive values indicate an increase in path length under the perturbed condition. The dashed lines indicate identity and the solid lines show the fitted regression.  $r^2$  and regression slope summarize sensitivity.

MediaPipe-derived path length demonstrated excellent agreement with the reference system (ICC = 0.985). Mean bias was small (1.25 mm) with 95% limits of agreement at approximately  $\pm 66$  mm. A slope

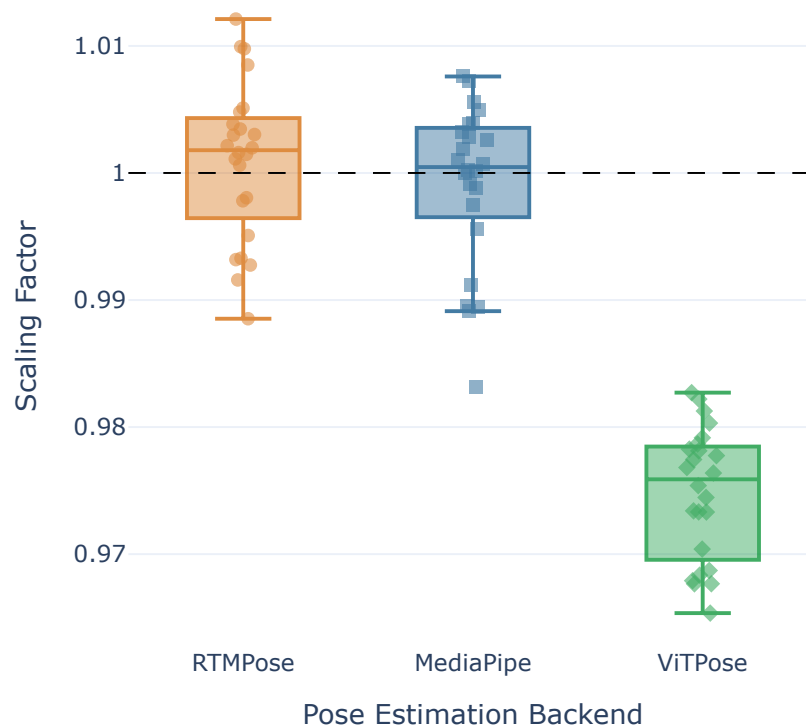
of 0.90 indicated proportional underestimation of the path length, and the condition-stratified Bland-Altman results showed that this underestimation became more apparent in the more challenging balance conditions. By comparison, ViTPose and RTMPose-derived data showed poor agreement ( $ICC < 0.10$ ), large positive biases of 726 and 1052 mm, respectively, and wide limits of agreement. Agreement statistics for all three backends are provided in Appendix 2—table 2, with identity and Bland-Altman plots for RTMPose and ViTPose shown in Appendix 2—figure 1.

MediaPipe also reproduced the condition-induced changes in COM path length. Across sensory perturbations, changes derived from MediaPipe were strongly associated with changes in the reference ( $r^2 = 0.83-0.96$ ), and regression slopes were close to the ideal value of one (0.89-1.06). This indicates that MediaPipe captured not only absolute differences among participants and trials, but also the magnitude of the postural response to altered visual and surface conditions. A slight underestimation was observed for the visual perturbation. In contrast, changes derived from ViTPose and RTMPose showed weak correspondence with the reference ( $r^2 = 0.01-0.33$ ), with slopes ranging from -5.48 to 1.45. Sensitivity statistics for all backends are provided in Appendix 2—table 3, and the corresponding RTMPose and ViTPose identity plots are shown in Appendix 2—figure 2.

COM velocity distributions provides an additional view of backend-specific measurement behavior (Figure 8—figure supplement 1). MediaPipe and the marker-based reference showed similar mediolateral and anteroposterior velocity distributions, although the MediaPipe distributions had consistently longer tails. Differences were larger in the vertical direction during solid-ground conditions. Because little true vertical COM movement is expected while standing on a firm surface, variability in vertical velocity provides an indication of measurement noise. In the Eyes Open/Solid Ground condition, the across-trial standard deviation was  $0.90 \pm 0.26$  mm/s for the reference,  $2.40 \pm 0.42$  mm/s for MediaPipe,  $9.60 \pm 1.94$  mm/s for ViTPose, and  $14.25 \pm 1.99$  mm/s for RTMPose.

### **Pose-estimation dependent scaling**

To evaluate global differences in reconstructed body size, we examined the uniform scale factor required to align each markerless reconstruction with the marker-based reference across trials from both tasks (Figure 9). A scale factor below 1 indicates that the markerless reconstruction was larger than the reference and therefore required downscaling. ViTPose showed a consistent scaling offset, with a median scale factor of 0.98 (range: 0.96-0.98), corresponding to reconstructions approximately 2-4% larger than the marker-based reference. By contrast, MediaPipe and RTMPose scale factors were centered near 1, indicating little systematic global scaling bias (MediaPipe median: 1.00, range: 0.98-1.01; RTMPose median: 1.00, range: 0.99-1.01).



**Figure 9.** Uniform scale factors estimated for each trial during spatial alignment of the markerless reconstructions to the marker-based reference. Values are shown for RTMPose (orange), MediaPipe (blue), and ViTPose (green). A scale factor below 1 indicates that the markerless reconstruction was larger than the reference and required downscaling. The dashed horizontal line indicates a scale factor of 1, corresponding to no global difference in reconstructed size.

[PLACEHOLDER: Maybe a video of all 3 fmc skeletons in their original alignment conditions? makes it very obvious where ViTPose is scaled larger]

## Discussion

In this study, we characterize a FreeMoCap motion capture setup using six low-cost webcams across two movement tasks: gait and balance. We found the 2D pose estimation backends used in this study, MediaPipe, RTMPose, and ViTPose, produced distinct error profiles, and their relative performance different across our movement tasks. Critically, the distinct behavioral differences show that validity of a markerless motion capture system is better understood in relation to a specific pipeline configuration and measurement context than as a fixed property of the system as a whole.

### Agreement across gait

Joint trajectory errors were generally under 30 mm and sagittal joint angle errors below 5°, consistent with ranges reported across other markerless validation studies (6-30 mm for trajectories, 2-11° for sagittal angles) (Nakano *et al.*, 2020; Kanko *et al.*, 2021; Ripic *et al.*, 2023; Uhlich *et al.*, 2023). These errors also fall within the 2-5° range that McGinley *et al.* characterize as potentially acceptable for many clinical interpretations, although suitability ultimately depends on the outcome and application (McGinley *et al.*, 2009).

Spatiotemporal gait parameters pooled across walking speeds showed minimal bias and strong agreement with the reference. Agreement was lowest for swing duration, although it remained strong ( $ICC \approx 0.90$ ), consistent with the reduced swing-phase agreement reported by Kanko et al. (Kanko *et al.*, 2021). A limitation of the present gait analysis is the 30 Hz camera sampling rate, which was intentionally retained to characterize performance using low-cost hardware. At this frame rate, gait event timing is resolved in approximately 33 ms increments, so a one frame difference represents a comparatively large proportion of the shorter swing phase. This limits sensitivity to small timing differences and may partly explain the lower agreement observed for swing duration. Future validation with higher framerate cameras could help distinguish pose-estimation error from temporal sampling limitations.

#### **Agreement and sensitivity across balance**

MediaPipe-derived COM path length preserved the progressive increase across conditions observed in the marker-based reference with strong agreement ( $ICC = 0.985$ ). MediaPipe-derived data also showed strong sensitivity to the effects of visual, proprioceptive, and combined visual/proprioceptive perturbations on balance, capturing condition-dependent changes in COM path length with near one-to-one agreement to the reference. Mean horizontal velocity was comparable between MediaPipe-derived data and the reference across all conditions, a notable finding given that sway velocity is regarded as one of the most reliable and clinically sensitive posturographic parameters (Jančová Všečeková and Drey, 2013; Paillard and Noé, 2015).

The present analysis characterizes balance using center of mass behavior, whereas conventional posturography more commonly relies on center of pressure (COP). Although COM and COP are biomechanically coupled (Winter, 2009) (i.e., COP moves to constrain the movement of COM in balance), they are not interchangeable measures. Panzer et al., for example, reported changes in eyes-closed balance that were detectable in COP but not COM (Panzer, Bandinelli and Hallett, 1995). Future work should therefore compare markerless COM estimates directly with force-plate COP and determine whether full-body kinematics provide complementary information beyond conventional posturographic measures.

#### **Pose estimation backends produce distinct downstream error profiles**

We found that each pose estimation backend produced distinct spatial and temporal characteristics that meaningfully affected the reconstructed 3D data and the biomechanical outcomes derived from it.

MediaPipe-derived COM trajectories were sufficiently stable to distinguish the small condition-dependent changes in postural sway that were not consistently recovered using RTMPose or ViTPose, both of which produced greater COM path length and velocity. However, the 95% ellipse area was similar across all three backends, suggesting that the overall spatial extent of sway was reasonably preserved even when path-dependent measures differed. One likely contributor to MediaPipe's stronger performance on path length and velocity is its internal temporal landmark smoothing, which uses an implementation of the One-Euro filter (Casiez, Roussel and Vogel, 2012). Because this filter applies stronger smoothing during slower movement and less during faster movement, it may suppress frame-to-frame jitter during quiet standing while preserving responsiveness during larger movements. Consistent with this interpretation, MediaPipe-derived hip and knee angles also showed speed-dependent differences during gait, with suprathreshold SPM clusters around peak excursions decreasing as walking speed increased. However, because filtered and unfiltered MediaPipe outputs were not directly compared, the contribution of temporal smoothing remains an interpretation rather than a demonstrated mechanism.

ViTPose-derived reconstructions showed a different spatial error profile. A median scaling factor of 0.98 (range: 0.96-0.98) was required for alignment with the marker-based reference, while trajectory error was disproportionately elevated in the vertical direction, suggesting that the discrepancy was not explained by uniform scaling alone. ViTPose-derived ankle angles also showed a persistent plantarflexed offset

and suprathreshold SPM clusters across much of the gait cycle. Because foot-segment orientation is particularly sensitive to ankle and toe landmark position, vertical landmark-positioning error may have contributed to this pattern. Hip and knee kinematics and anteroposterior gait measures were nevertheless generally accurate, indicating that ViTPose errors were outcome dependent rather than uniformly distributed across the reconstruction.

RTMPose-derived reconstructions were comparatively consistent across spatial, kinematic, and spatiotemporal gait outcomes and produced the lowest ankle-angle errors among the evaluated backends. In contrast, RTMPose-derived COM trajectories did not consistently distinguish among the balance conditions, illustrating that strong performance during large-amplitude gait movements does not necessarily imply sufficient temporal stability for resolving subtle postural displacements.

Differences between pose estimators are often attributed primarily to their training datasets, but the present findings suggest that architectural and implementation-level choices may also shape downstream measurements in important ways. MediaPipe’s stronger balance performance, for example, may reflect its use of adaptive temporal smoothing rather than an inherent advantage of the underlying pose estimator itself. If so, other backends such as RTMPose or ViTPose may also support sensitive balance measurements when paired with appropriate temporal filtering. More broadly, pose-estimation methods should be characterized not only by overall accuracy, but by the spatial and temporal properties they introduce into reconstructed trajectories and by how those properties interact with the outcome being measured. These implementation details should therefore be documented explicitly in validation studies and treated as potentially modifiable components of the analysis pipeline.

### **Markerless motion capture validity depends on movement context**

Markerless motion capture performance varied across movements with different spatial and temporal demands, and performance in one context did not necessarily predict performance in another. Meaningful variation was evident even within gait: as walking speed increased, joint-position and joint angle errors generally increased, particularly in the anteroposterior and vertical directions at distal joints. These effects were not uniform across joints, axes, outcomes, or pose estimation backends, indicating that movement speed interacted with specific characteristics of the measurement pipeline rather than simply increasing error globally. The range of walking speeds examined here was therefore useful not only for evaluating gait across realistic conditions, but also for revealing error patterns that may have remained hidden in a single-speed validation.

More broadly, performance established under one version of a task should not be assumed to generalize across the full range of conditions in which that task may be performed. Markerless motion-capture systems may behave differently as movement speed, amplitude, complexity, viewpoint, or environmental conditions change. Validation is therefore most informative when it contributes to a profile of system performance across representative conditions rather than reducing the system to a single estimate of accuracy.

### **Building richer profiles of markerless motion capture performance**

The present findings suggest that markerless motion capture benchmarking should focus on characterizing the spatial and temporal behavior of specific implementations in relation to the movement and outcome of interest. Rather than asking which pose estimation algorithm is most accurate overall, researchers may benefit more from knowing which algorithms are most suitable for the specific movements and conditions they are interested in exploring.

Developing these performance profiles requires validation methods that are themselves transparent and reusable. If markerless motion-capture systems are expected to operate across laboratories, clinics, community settings, and other real-world environments, their behavior will need to be characterized

under variation in factors such as camera quality, frame rate, lighting, camera number and configuration, participant characteristics, and movement type. To support this process, we have released the validation pipeline used to generate the analyses reported here so that the same procedures can be reproduced, modified, and extended. We plan to expand this work toward a broader set of benchmarking tools for systematically characterizing pose-estimation and reconstruction behavior across tasks and acquisition conditions.

This approach may be especially important as markerless motion capture continues to become more compositional. The ability to exchange pose estimation backends means that an algorithm need not be treated as a fixed property of a motion capture system. Instead, pose estimators can be selected according to the movement, population, or outcome of interest and evaluated within a common acquisition and reconstruction framework. FreeMoCap is being developed to support this model by providing standardized interfaces through which alternative pose estimation methods can be integrated, configured, evaluated, and made available to other users.

### **Practical use and broader application of FreeMoCap**

FreeMoCap has already been applied across a range of research and applied contexts, including gait and other biomechanical analyses, reaching and grasping (Mazrouei *et al.*, 2026), computational linguistics (Orr, Peechatt and Alm, 2024; Peechatt, Alm and Bailey, 2024; Nunnari *et al.*, 2024), animation (Landa *et al.*, 2025), and robotic or assistive-device development (Obukhov *et al.*, 2026). This breadth reflects one advantage of retaining access to reconstructed three-dimensional landmark trajectories rather than constraining the system to a single downstream biomechanical model or analysis.

A limitation of the present study is that data were collected under controlled laboratory conditions that were favorable for markerless reconstruction. The recording space was well lit and relatively uncluttered, camera placement was unconstrained, and sufficient room was available to deploy all six cameras around the participant. These conditions likely reduced challenges associated with occlusion, poor illumination, restricted viewpoints, and limited camera placement that may be encountered in clinical, community, or home environments. The present results therefore establish performance under an intentionally low-cost hardware configuration, but not under the full range of environmental constraints that may accompany real-world deployment.

More generally, accessibility does not remove practical measurement constraints. Consumer cameras introduce limits in frame rate, image quality, synchronization, field of view, and sensitivity to environmental conditions, while markerless reconstruction remains dependent on appropriate camera placement, calibration, visibility, and the quality of the pose-estimation output. Practical guidance for acquisition, calibration, camera configuration, and interpretation of reconstructed data is therefore an important complement to quantitative validation. We provide additional discussion of these considerations in [Appendix/Supplementary Note X].

Taken together, the present results support viewing FreeMoCap not only as a low-cost motion-capture system, but as an open framework for progressively characterizing how markerless measurement behaves across algorithms, tasks, and acquisition conditions. The goal is not to identify a single configuration that is universally optimal, but to make the behavior and limitations of different configurations sufficiently visible that users can select and evaluate methods appropriate to their particular application.

### **Conclusion**

A low-cost, six-camera FreeMoCap system produced meaningful gait measurements across three pose estimation backends and, when using MediaPipe, detected small condition-dependent changes in postural sway. However, performance varied across backends, outcomes, tasks, and walking speeds. These results

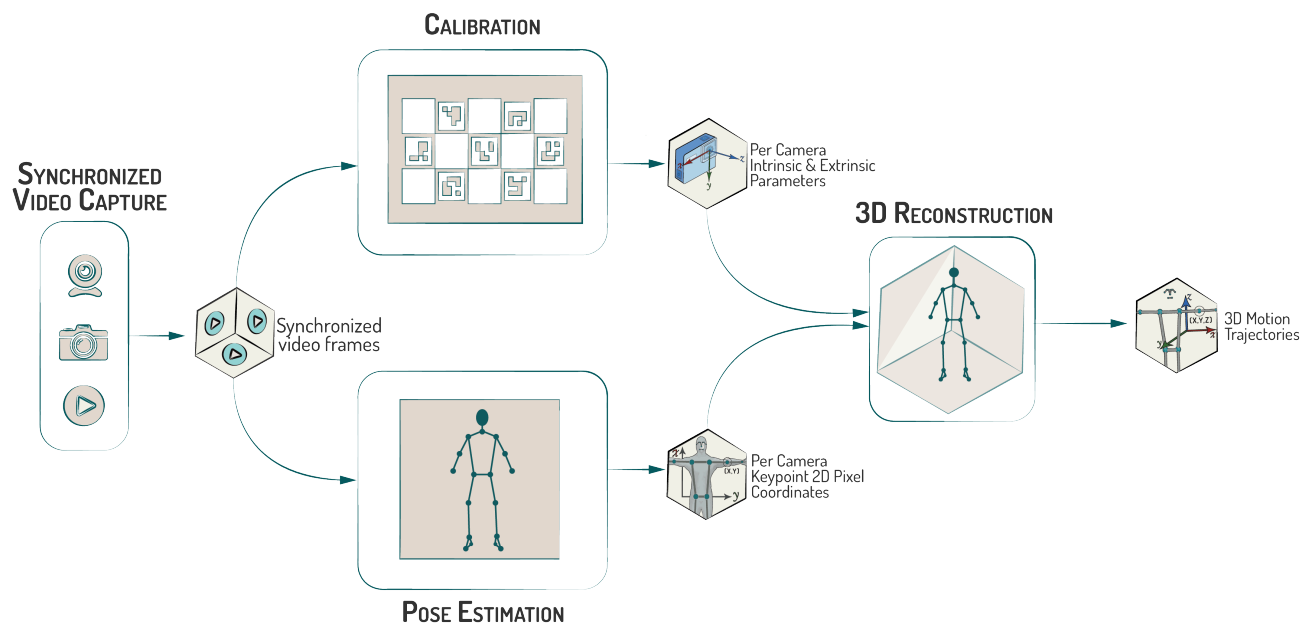
show that markerless motion capture validity is conditional on the complete measurement configuration and the movement context in which it is evaluated, rather than being a fixed property of a software platform or pose estimation model. Validation and benchmarking should therefore characterize the spatial and temporal behavior of specific implementations across representative movement conditions and clearly report the boundaries within which acceptable performance has been demonstrated.

## Materials and Methods

### FreeMoCap Software

FreeMoCap (Queen *et al.*, 2024) is a fully open-source markerless motion capture framework designed to maximize accessibility across the entire workflow. It operates with consumer-grade webcams, requires neither physical markers nor a specialized recording environment, and supports the full process from synchronized video acquisition to 3D kinematic reconstruction (Figure 10). Its modular, tracker-agnostic architecture allows different pose estimation backends to be incorporated according to the needs of a given application.

FreeMoCap is organized as a polyrepo, with major components of the motion capture workflow maintained in separate repositories. The following sections describe the principal stages of the system: synchronized video acquisition, camera calibration, 2D pose estimation, and 3D reconstruction.



**Figure 10.** PLACEHOLDER - Figure needs adjustment. Maybe add some text to it? seems a bit empty.

### Synchronized Video Acquisition

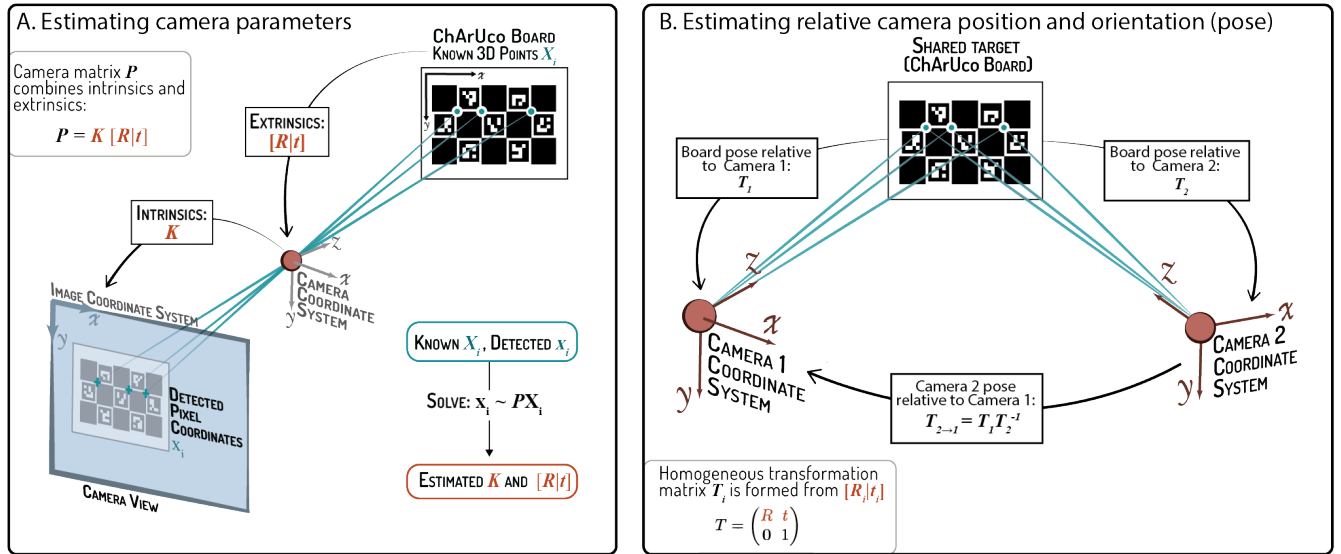
Synchronous video recording refers to the acquisition and temporal alignment of video streams. For accurate 3D reconstruction, each set of frames must correspond to the same moment in time. Without proper synchronization, time lags between cameras can result in inaccurate 3D data.

In FreeMoCap, synchronous recording is handled primarily by SkellyCam, a software package to provide high quality synchronous recording methods that enable the use of low-cost hardware - specifically

consumer-grade, off-the-shelf webcams. However, these cameras may not be suitable for all research needs (e.g., capturing athletic performance which may necessitate higher frame rate cameras or outdoor recordings, which may require non-USB cameras). The software accommodates these use cases in two ways: 1) Videos collected from a set of external cameras (e.g., GoPros or smartphone cameras) can be synchronized within the main software using light and audio-based synchronization methods; 2) A pre-synchronized set of videos can be directly imported into the software for processing. We aim to reduce dependency on specific hardware configurations and allow data collection protocols to be adapted to the needs of the study, rather than constrained by the system itself.

### Camera Calibration

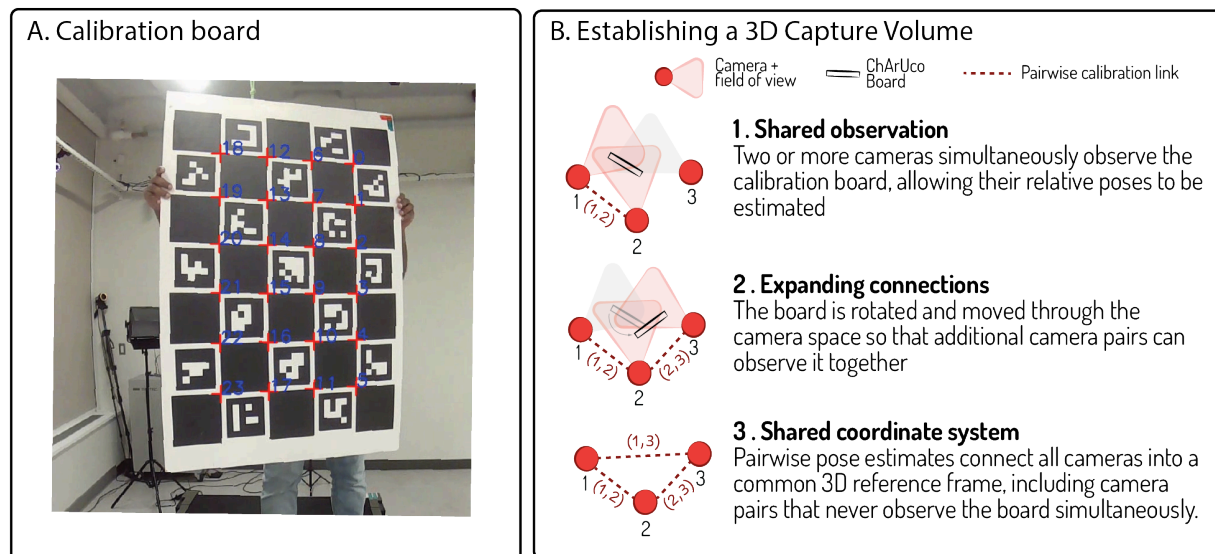
In order to reconstruct 3D data from the 2D camera images, it is necessary to determine how each camera observes the world and where it is positioned within it. The former describes intrinsic parameters (i.e., how each camera maps light onto its image sensor). The latter describes extrinsic parameters (i.e., the position and orientation of each camera relative to a common world reference frame). Camera calibration is the process of calculating the intrinsic and extrinsic parameters of a set of cameras and establishing the geometric relationships required to project 2D image features into a shared 3D coordinate system (Figure 11).



**Figure 11.** **A.** Estimation of camera intrinsic and extrinsic parameters using a ChArUco calibration board. Each ArUco marker has a unique identifier, allowing the intervening chessboard corners to be detected and assigned known locations within the board coordinate system. The board origin is defined at the first ChArUco corner, and the 3D coordinates of the remaining corners are determined from the known printed square size. Across multiple views of the board, correspondences between these known 3D corner locations, ( $X_i$ ), and their detected 2D pixel coordinates, ( $x_i$ ), are used to estimate the camera intrinsics, ( $K$ ), and the board-to-camera extrinsics, ( $[R|t]$ ). Together, these form the camera projection matrix ( $P = K[R|t]$ ). **B.** Estimation of each camera’s position and orientation, collectively referred to as its pose, within a shared coordinate system. The extrinsic parameters estimated for each camera are first written as homogeneous transformation matrices,  $T_i$ . Camera 1 is then selected as the reference frame, and the pose of Camera 2 relative to Camera 1 is obtained by composing the two transforms:  $T_{2 \rightarrow 1} = T_1 T_2^{-1}$ . Repeating this process for the remaining cameras establishes the geometric relationships among all cameras in the system.

Calibration and 3D reconstruction are implemented in FreeMoCap using a modified implementation of the Anipose toolkit (Karashchuk *et al.*, 2021). The calibration tool is a ChArUco board, a hybrid calibration target consisting of a checkerboard pattern overlaid with uniquely identifiable ArUco markers. This board can be printed on standard paper and mounted to a rigid surface, or printed directly onto a rigid board.

During calibration, the ChArUco board is moved throughout the intended capture volume and presented to multiple cameras simultaneously. Each shared observation establishes a geometric relationship between the cameras that can see the board. By moving the board through different regions of the capture space, additional overlapping camera pairs are linked together, forming a connected calibration network (Figure 12). Consequently, two cameras that never observe the board at the same time can still be related through one or more intermediate cameras. As long as all cameras belong to the same connected network, their poses can be expressed within a common 3D coordinate system.



**Figure 12.** **A.** A ChArUco calibration board detected in a single video frame. Each ArUco marker has a unique identifier, allowing the intervening ChArUco corners to be detected and matched across images. The detected corner IDs are shown in blue. **B.** Establishing a shared 3D capture volume. When two or more cameras simultaneously observe the calibration board, their relative poses can be estimated. Moving and rotating the board through the capture space creates additional pairwise calibration links, progressively connecting all cameras into a common 3D reference frame, including camera pairs that do not directly observe the board at the same time.

[PLACEHOLDER: Maybe link a video of an example calibration here from the data collection?]

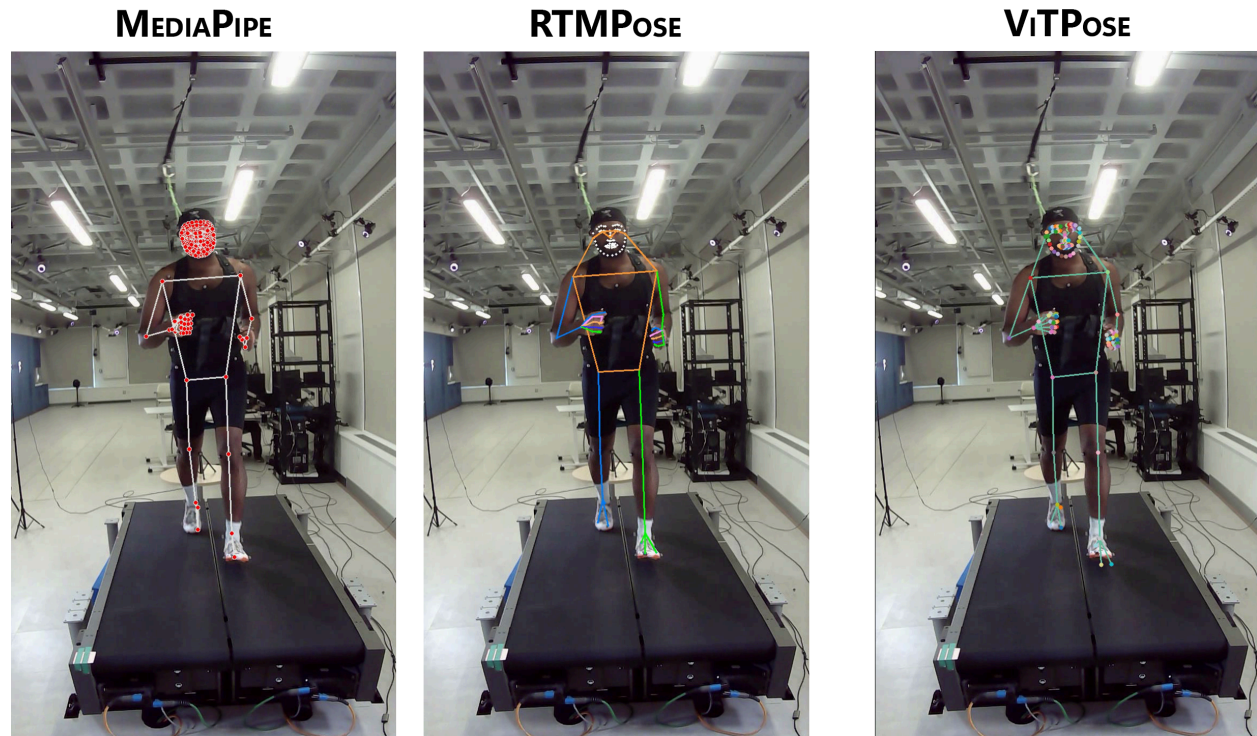
The board should be observed across a range of positions and orientations. In particular, variation in depth and tilt provides stronger constraints for estimating camera intrinsic and extrinsic parameters than observations confined to a single plane or orientation. The board must remain rigid, as bending or flexing changes the assumed geometry of the calibration target and can introduce parameter-estimation errors. Glare should also be minimized because reflections can obscure markers and corners; we therefore recommend printing the target on matte, non-glossy material.

At the start of a recording, the ChArUco board may be placed flat on the floor within the shared field of view of the cameras, a procedure we refer to as ground-plane calibration. The detected board pose is then used to define the reconstruction coordinate system: the plane of the board is assigned to  $Z = 0$ , and the coordinate axes are aligned with the board orientation. As a result, reconstructed 3D kinematic data are expressed directly in a physically meaningful, ground-aligned reference frame, reducing the need for post hoc translation or rotation.

### Pose Estimation

Pose estimation converts each camera’s video frames into 2D anatomical keypoints for subsequent 3D reconstruction (Figure 13). The default FreeMoCap pipeline uses MediaPipe (Lugaresi *et al.*, 2019), a free and open-source framework incorporating the BlazePose convolutional neural network (Bazarevsky *et al.*,

2020). MediaPipe was selected as the default backend primarily for its accessibility: it is straightforward to install and use through Python, is computationally lightweight, and can run without a dedicated GPU. These characteristics allow the complete FreeMoCap pipeline to operate on a broad range of consumer hardware.



**Figure 13.** Examples of landmarks estimated on a human body from three different pose estimation algorithms. **Left:** MediaPipe; **Middle:** RTMPose; **Right:** ViTPose.

However, the choice of pose estimation model can substantially affect the resulting motion-capture data. Errors in 2D keypoint localization propagate into reconstructed 3D trajectories and derived kinematic measures, and performance varies across pose estimation algorithms (Needham *et al.*, 2021; Washabaugh *et al.*, 2022; Ceriola *et al.*, 2024). Moreover, many general-purpose models are trained on datasets that were not designed specifically for movement-science applications and may provide limited representation of particular movements, environments, or populations (Seethapathi *et al.*, 2019; Needham *et al.*, 2021). No single pose estimation model is therefore likely to be optimal across all applications.

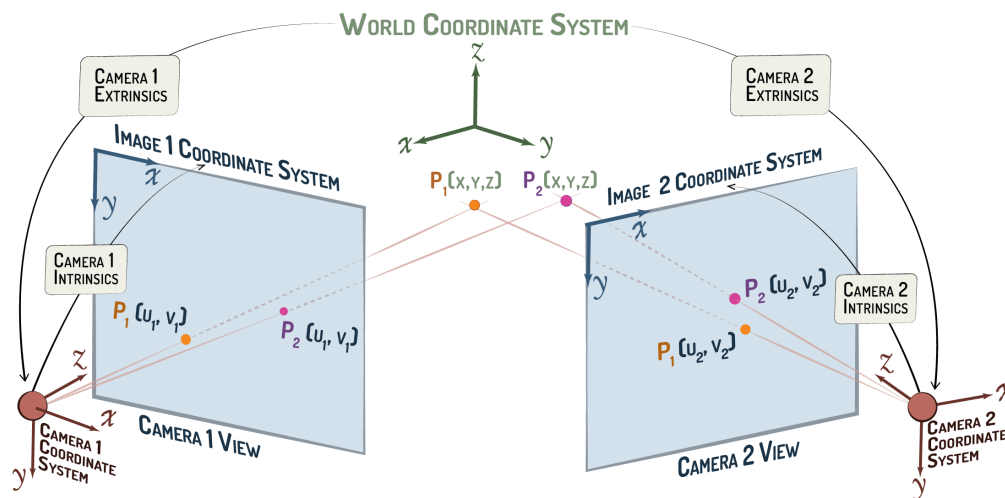
To accommodate alternative models, FreeMoCap separates pose estimation from the remainder of the processing pipeline through SkellyTracker, its pose estimation management framework. SkellyTracker defines a standardized interface through which a model receives video frames and returns keypoint coordinates in a consistent format. New pose estimation backends can therefore be added by implementing this interface, without requiring corresponding changes to camera calibration, 3D reconstruction, post-processing, or data export. This avoids the need to construct a new motion-capture pipeline whenever a different pose estimation model is required.

This modular design also supports controlled comparisons among pose estimation algorithms. Different backends can be applied to the same synchronized videos while holding the camera configuration, calibration, reconstruction, and post-processing procedures constant. Differences in the resulting 3D estimates can therefore be more directly attributed to the pose estimation stage, supporting the systematic

benchmarking of markerless pose estimation algorithms identified as a need within the movement-science community (Needham *et al.*, 2021).

### 3D Reconstruction

The 2D keypoints detected independently in each camera view are combined to reconstruct their positions in three dimensions (Figure 14). For each frame and keypoint, the detected pixel coordinate in each camera defines a line of sight extending from that camera into the calibrated capture volume. The corresponding 3D position is estimated by triangulating across these observations using the camera projection matrices obtained during calibration. FreeMoCap performs this reconstruction using direct linear transformation (DLT) through a modified implementation of the Anipose toolkit (Karashchuk *et al.*, 2021).



**Figure 14.** 3D reconstruction through multi-view triangulation. Each camera observes the same anatomical keypoints, shown here as  $P_1$  and  $P_2$ , at different 2D pixel locations. Together with the calibrated camera projection matrices, each detection defines a line of sight from the camera into the shared 3D coordinate system. The position of each keypoint is estimated from the convergence of the corresponding lines of sight across camera views.

Corresponding 2D keypoints from the synchronized camera views were triangulated into 3D coordinates using the calibrated camera projection matrices and direct linear transformation.

In multiview markerless motion capture, a camera view that is generally informative may nevertheless produce erroneous 2D detections during particular movements or periods of occlusion. These localized errors can substantially degrade the triangulated 3D trajectory. One approach is to exclude the affected camera from the entire recording; however, this also removes the many valid observations contributed by that camera at other frames and keypoints. To retain these usable observations, we implemented an optional progressive outlier-rejection procedure that excluded individual camera observations locally during triangulation rather than removing a camera view globally.

For each keypoint and frame, an initial 3D position was estimated using all available camera views, and the mean reprojection error was calculated. When this error exceeded a specified threshold, the keypoint was retriangulated using each possible subset formed by omitting one camera. The leave-one-camera-out solution with the lowest mean reprojection error was compared with the original all-camera solution. When excluding one camera produced a sufficiently large reduction in reprojection error, the refined

estimate was used; for intermediate improvements, the original and refined estimates were blended to reduce abrupt transitions between reconstruction solutions.

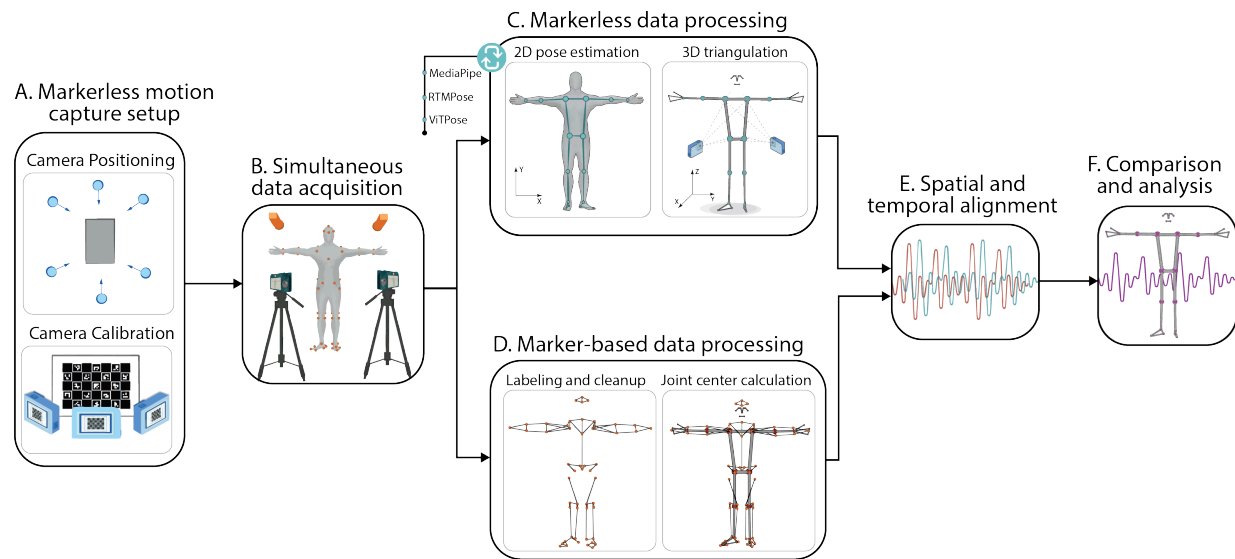
Outlier rejection was enabled only for recordings in which the default reconstruction contained substantial artifacts attributable to a camera view. Each recording's default (all-camera) reconstruction was visually inspected alongside the annotated 2D videos from each camera view. The procedure was enabled when artifacts in the 3D trajectories (e.g., abrupt spatial discontinuities or anatomically implausible excursions) could be traced to visibly erroneous 2D keypoint estimates in a specific camera view, rather than being attributed to 2D estimation failure by inference alone. This determination was based solely on the quality of the markerless reconstruction and was made independently of agreement with the marker-based reference.

When enabled for a recording, the same procedure was applied to MediaPipe, RTMPose, and ViTPose reconstructions. In these cases, the procedure recovered usable reconstructions that would otherwise have required exclusion while preserving valid observations from the affected camera views.

The reconstructed trajectories were subsequently processed using SkellyForge, FreeMoCap's post-processing package. Short gaps are interpolated when no acceptable triangulation solution is available for a frame, after which the coordinate trajectories are low-pass filtered using a Butterworth filter to attenuate high-frequency noise. The resulting data consist of temporally continuous 3D keypoint trajectories expressed in the shared coordinate system established during calibration.

#### **Validation Procedure**

A broad overview of the procedure is found in Figure 15.



**Figure 15.** Overview of experimental design and data processing methods. **A.** Six generic USB webcams were placed circularly around the treadmill and connected to a single PC, with the FreeMoCap software used for video acquisition. The cameras were calibrated using a ChArUco board measuring 1016 x 698.5 mm (40 x 27.5 in), with a square size of 126 mm. **B.** Participants completed two trials each of the gait and balance assessments while being recorded simultaneously by the markerless and marker-based motion capture systems. **C.** Markerless video data were processed using three pose estimation backends: MediaPipe, RTMPose, and ViTPose. The resulting 2D keypoint detections were triangulated into 3D joint-center trajectories using the FreeMoCap pipeline. **D.** Marker-based trajectories were labeled and cleaned, and anatomical joint centers were calculated from the marker trajectories. **E.** Markerless and marker-based trajectories were temporally aligned using cross-correlation and transformed into a common spatial coordinate system. **F.** The aligned markerless and marker-based joint-center trajectories were used for task-specific comparisons and statistical analyses.

### Participants

6 participants (5 male, 1 female) were recruited in this study. All participants were healthy adults with no reported musculoskeletal or neurological impairments affecting gait. The participants provided informed consent and the protocol was approved by the Northeastern University IRB (#19-08-27).

### Experimental Setup

#### Marker-based motion capture

The marker-based system consisted of 9 Miquis M3 and 2 Oqus 700+ cameras (300 Hz) and tracked the positions of 48 markers. Four markers were affixed to a head cap (top, front, left, right). An additional marker was placed on the C7 vertebra. Trunk markers were placed on the sternum, right lower back, and bilaterally on the acromion processes and on the anterior and posterior aspects of the shoulders. Upper extremity markers were placed bilaterally on the medial and lateral humeral epicondyles, the ulnar and radial styloid processes, and dorsum of the hands. Pelvic markers were placed bilaterally on the anterior superior iliac spines (ASIS), iliac crests, and greater trochanters. Posterior pelvic motion was tracked using a rigid sacral plate containing markers positioned over the left and right posterior superior iliac spines (PSIS) and the sacrum. Lower extremity markers were placed bilaterally on the medial and lateral femoral epicondyles, medial and lateral malleoli, calcanei, 1st and 5th metatarsal heads, and dorsal aspect of the second metatarsal.

### **Markerless motion capture**

The markerless system consisted of six consumer-grade cameras (\$20 USB webcams, 1280x720 resolution, 30Hz) arranged in a circle around the capture volume. Cameras were positioned to maximize multi-view coverage of the participant. Two cameras were aligned with the frontal plane, positioned anterior and posterior to the participant. The remaining four cameras were placed at approximately 45° oblique angles relative to the sagittal plane. Camera positions were standardized across participants using floor markers, while camera height and orientation were adjusted for each participant to optimize visibility and reduce occlusion.

All cameras were connected to a single acquisition computer. Video capture was performed using the FreeMoCap software. Because both FreeMoCap and Qualisys recordings were acquired on the same computer, timestamps from each system were referenced to a shared system clock, enabling temporal alignment between datasets.

Prior to recording, cameras were calibrated using a ChArUco calibration board (square size: 126 mm, board size: 1016 x 698.5 mm). At the start of each recording, the board was placed flat on the floor within view of all cameras to define the reference frame of the reconstruction (ground plane alignment). The origin of the capture volume was set using the board, and the reconstructed 3D data were aligned such that the vertical axis corresponded to the Z-axis, with the horizontal plane defined as  $Z = 0$ .

### **Data Collection**

Each participant completed two trials of both the gait and balance tasks. At the start of each trial, participants assumed an “A-pose”, a neutral position with feet slightly apart, head up, and arms angled downward at roughly 45 degrees for a few seconds.

### **Gait**

Participants walked on a treadmill at progressively increasing speeds. The treadmill increased in 0.50 m/s increments every 30 seconds, starting from rest until 2.50 m/s. Prior to data collection, participants were given time to familiarize themselves with the treadmill and each speed condition.

### **Balance**

Participants were asked to complete the Modified Clinical Test of Sensory Interaction on Balance (CTSIB-M) (*Modified Clinical Test of Sensory Interaction on Balance | RehabMeasures Database*, 2013). This test consists of four 60-second conditions that vary participant visual condition (i.e., standing with eyes open vs. eyes closed) and standing surface (i.e., standing on a solid surface vs. a foam pad; ProsourceFit Exercise Balance Pad): (1) Eyes Open/Solid Ground; (2) Eyes Closed/Solid Ground; (3) Eyes Open/Foam Pad; (4) Eyes Closed/Foam Pad. Participants stood with hands at their sides. To minimize camera occlusions, participants were asked to place feet shoulder-width apart. This placement has been shown to have little effect on CTSIB-M scoring (Wrisley and Whitney, 2004).

The CTSIB-M was selected because: 1) Static balance tasks present minimal body movement, providing a method of measuring system sensitivity to small-amplitude kinematics; 2) The graded condition structure allows evaluation of whether particular postural challenges affect system accuracy.

Participants were given a visual fixation target during eyes-open conditions and a floor mark to standardize starting position. Participants completed two trials of the full protocol, each recorded simultaneously by both the marker-based and markerless motion capture systems.

## **Data Processing**

### **Marker-based motion capture data**

Marker-based data were tracked, labeled and processed in QTM. Missing trajectories were interpolated using linear or relational gap-filling. The labeled and cleaned marker positions were exported as a .tsv file containing system timestamps. Head, elbow, wrist, knee and ankle joint centers were defined as the midpoint of their respective medial and lateral markers. Shoulder joint centers were defined as the midpoint of the anterior and posterior markers. The hip joint centers were estimated using the methods described by Bell et al. (Bell, Brand and Pedersen, 1989). Raw kinematic data were filtered using a zero-lag, fourth-order Butterworth filter with a 6 Hz cutoff frequency.

### **Markerless motion capture data**

#### *Pose estimation*

Synchronized videos were processed using the FreeMoCap pipeline (v1.7.4). Two-dimensional keypoints were estimated using MediaPipe (Lugaresi *et al.*, 2019) (mediapipe v0.10.14), RTMPose (Jiang *et al.*, 2023) (rtmlib v0.0.14), and ViTPose (Xu *et al.*, 2022), implemented using the easy\_ViTPose repository. RTMPose was run using its WholeBody configuration (mode = performance), with RTMDet used for person detection. ViTPose used the ViTPose-H WholeBody model (vitpose-h-wholebody.pth), with YOLOv8 medium (yolov8m.pt) used to detect person bounding boxes in every frame. Both models produced the COCO-WholeBody 133-keypoint topology. MediaPipe was implemented using the Holistic solution with the heavy pose model. MediaPipe Holistic jointly estimated body, hand, and facial landmarks. Although all three backends produced keypoints beyond those required for the present study, only body keypoints were retained for subsequent reconstruction and analysis.

The backends also differed in their temporal processing. RTMPose and ViTPose estimated poses independently in each frame, whereas MediaPipe was run in video mode (static\_image\_mode = false) with landmark smoothing enabled (smooth\_landmarks = true). In the legacy MediaPipe Holistic solution, temporal landmark smoothing is conditional on both settings: it is disabled either when static-image mode is enabled or when landmark smoothing is turned off. Thus, settings that may appear to be minor implementation details can alter the temporal characteristics of the resulting trajectories and, consequently, their validation against a reference system. Complete model and inference configurations for all three pose-estimation backends are therefore reported in Appendix 3—table 1.

#### *Triangulation and post-processing*

Corresponding 2D keypoints from the synchronized camera views were triangulated into 3D coordinates using the calibrated camera projection matrices and direct linear transformation.

In multiview markerless motion capture, a camera view that is generally informative may nevertheless produce erroneous 2D detections during particular movements or periods of occlusion. These localized errors can substantially degrade the triangulated 3D trajectory. One approach is to exclude the affected camera from the entire recording; however, this also removes the many valid observations contributed by that camera at other frames and keypoints. To retain these usable observations, we implemented an optional progressive outlier-rejection procedure that excluded individual camera observations locally during triangulation rather than removing a camera view globally.

For each keypoint and frame, an initial 3D position was estimated using all available camera views, and the mean reprojection error was calculated. When this error exceeded a specified threshold, the keypoint was retriangulated using each possible subset formed by omitting one camera. The leave-one-camera-out solution with the lowest mean reprojection error was compared with the original all-camera solution.

When excluding one camera produced a sufficiently large reduction in reprojection error, the refined estimate was used; for intermediate improvements, the original and refined estimates were blended to reduce abrupt transitions between reconstruction solutions.

Outlier rejection was enabled only for recordings whose default reconstruction contained substantial 3D artifacts. Each recording's default (all-camera) reconstruction was visually inspected alongside the annotated 2D videos from each camera view. The procedure was enabled when artifacts in the 3D trajectories (e.g., abrupt spatial discontinuities or anatomically implausible excursions) could be traced to visibly erroneous 2D keypoint estimates in a specific camera view, rather than being attributed to 2D estimation failure by inference alone. This determination was based solely on the quality of the markerless reconstruction and was made independently of agreement with the marker-based reference.

When enabled for a recording, the same procedure was applied to MediaPipe, RTMPose, and ViTPose reconstructions. In these cases, the procedure recovered usable reconstructions that would otherwise have required exclusion while preserving valid observations from the affected camera views.

Following reconstruction, gaps in the 3D trajectories were interpolated, and the trajectories were low-pass filtered using a zero-phase, fourth-order Butterworth filter with a cutoff frequency of 6 Hz.

### **Data synchronization and alignment**

Joint center trajectories from marker-based and markerless systems were temporally aligned using recorded Unix timestamps from both systems, which were generated on the same acquisition computer. Marker-based data were resampled to match the markerless sampling rate (30 Hz). Residual temporal offsets were further refined using cross-correlation of joint trajectories, followed by manual inspection.

Markerless data were spatially aligned to the marker-based reference frame using a least-squares optimized rigid transformation that minimized joint center errors between systems. The transformation consisted of three rotational ( $r_x, r_y, r_z$ ) and three translation ( $t_x, t_y, t_z$ ) parameters.

To identify a transformation that was consistent over the full recording, candidate transformations were estimated from randomly sampled subsets of frames and evaluated across the entire dataset. The transformation that minimized the global joint center error across all frames was selected for each trial.

### ***Data Analysis: Gait***

#### **Joint angles**

Joint angles were calculated as the Cardan XYZ decomposition of the relative rotation between adjacent segments. Sagittal-plane lower-body kinematics were extracted and analyzed across gait cycle-normalized strides. Joint angles were offset-corrected by subtracting the mean angle measured during the neutral A-pose stance at the start of each trial.

#### **Gait event detection**

Heel strike and toe off events were identified based on anteroposterior velocity zero-crossings of the foot, using the methods described by Zeni et al. (Zeni, Richards and Higginson, 2008). Marker-based gait events were used as the reference for time-normalizing to 0-100% of the gait cycle for all pose-estimated derived trajectories and joint angles.

#### **Gait parameters**

Spatiotemporal parameters were calculated for each system using their respective gait events. The following gait parameters were calculated: [UPDATE]

1) **Stance duration:** The time from **heel strike to subsequent toe off** of the same foot in milliseconds (ms).

2) **Swing duration:** The time from **toe off to subsequent heel strike** of the same foot in milliseconds (ms).

3) **Stride duration:** The time from **heel strike to subsequent heel strike** of the same foot in milliseconds (ms).

4) **Step length:** The **anteroposterior distance between the contralateral ankle at contralateral heel strike and the ipsilateral ankle at the subsequent ipsilateral heel strike**, reported in millimeters (mm). To account for the treadmill, step length was corrected using the distance traveled by the belt. Distance traveled by the belt was computed using the treadmill speed and the time between heel strikes.

5) **Stride length:** The **anteroposterior distance between the ipsilateral ankle at ipsilateral heel strike and the same joint at the subsequent ipsilateral heel strike**, reported in millimeters (mm). To account for the treadmill, stride length was corrected using the distance traveled by the belt. Distance traveled by the belt was computed using the treadmill speed and the time between heel strikes.

## Statistical analyses

Statistical analyses were performed using Python v3.11. Root mean squared error (RMSE) was calculated across all gait cycle-normalized joint center trajectories and joint angles. Per-trial RMSE was obtained by averaging across strides within a trial, and mean  $\pm$  SD were then computed across all trials.

To identify regions of significant difference between the marker-based reference and each pose estimation backend, statistical parametric mapping (SPM) two-tailed paired t-tests were performed on gait cycle-normalized joint angles using the `spm1d` package. SPM extends hypothesis testing to an entire timeseries, identifying continuous regions where differences exceed a critical threshold. SPM $\{t\}$  statistics were computed across the gait cycle, and statistical significance was assessed at  $\alpha = 0.05$ .

For each gait parameter, Bland-Altman plots with bias and 95% limits of agreement (LOA) were created (Bland and Altman, 1986). Intraclass correlation coefficients (ICC(2,1)) were calculated using the `pingouin` package to assess agreement (Shrout and Fleiss, 1979). ICC values under 0.5 were interpreted as poor agreement, 0.5-0.75 interpreted as moderate agreement, 0.75-0.90 as good agreement, and greater than 0.90 as excellent agreement (Koo and Li, 2016). Bland-Altman and ICC values were calculated across all speeds as well as per walking speed.

## Data Analysis: Balance

### Center of mass (COM) calculation

For the reference system and each pose estimation backend, body segments were defined with segment mass fractions and COM locations according to the anthropometric tables reported by Winter (Winter, 2009). Anthropometric data were used to calculate segment COM position. Total-body COM was calculated as the mass-weighted average of all segment COM coordinates.

### Center of mass path length

Using a custom-built viewer (available at: [https://github.com/aaroncherian/nih\\_balance\\_analyses](https://github.com/aaroncherian/nih_balance_analyses)), each trial was annotated with the start and stop frame numbers for each of the four conditions. 1600 frames were analyzed per standing condition within a trial. Within each condition, COM path length was calculated as the cumulative sum of the Euclidean distance between consecutive 3D COM positions.

## Postural metrics

The 95% confidence ellipse area, a measure often used in posturography, is the area that is expected to enclose approximately 95% of the points on the COM path (Prieto *et al.*, 1996). The ellipse axes were

derived from the decomposition of the 2D position covariance matrix. Semi-axis lengths were scaled to enclose 95% of the data. The scaling factor ( $\chi^2_{0.95,2} \approx 5.991$ ) is derived from the chi-squared distribution with two degrees of freedom (Schubert and Kirchner, 2014). Ellipse area was then calculated as  $\pi ab$ , where  $a$  and  $b$  are the scaled semi-axis lengths.

COM velocity was calculated per axis as the frame-to-frame displacement of COM position divided by the sampling interval. Mean 2D COM velocity in the horizontal plane was then calculated as the trial-level mean of the magnitude of the mediolateral and anteroposterior velocity components.

## Noise characterization

During quiet standing, vertical COM displacement is expected to be minimal, making vertical velocity a useful indicator of measurement noise. Trial-wise standard deviations of vertical COM velocity were calculated for the Eyes Open/Solid Ground condition and summarized across participants (mean  $\pm$  SD) to characterize the baseline noise floor of each pose estimation backend and the marker-based reference system under minimal-movement conditions.

## Sensitivity analysis

For markerless and marker-based systems, we calculated within-participant COM path length differences between conditions representing specific perturbations: 1) **Visual perturbation** using the path length difference between the Eyes Open/Solid Ground and Eyes Closed/Solid Ground conditions; 2) **Proprioceptive perturbation** using the path length difference between Eyes Open/Solid Ground and Eyes Open/Foam conditions; 3) **Combined visual/proprioceptive perturbation** using the path length difference between Eyes Open/Solid Ground and Eyes Closed/Foam conditions. For each perturbation, markerless path length differences were compared to marker-based path length differences using linear regression, and we report slope alongside  $r^2$ .

## Statistical analyses

To assess accuracy between the markerless and marker-based reference systems, we calculated Bland-Altman statistics with bias and 95% limits of agreement (LoA) alongside intraclass correlation coefficients (ICC (2,1)), using the packages and interpretations described above.

## Data Analysis: Scaling

To assess whether systematic scaling differences were present between pose estimation outputs and the marker-based reference, we extended our spatial optimization to include an additional uniform scaling parameter ( $s$ ). In this formulation, the transformation consisted of rotation, translation, and a single global scale factor applied isotropically across all spatial dimensions.

For each trial, the optimal scale factor was estimated alongside the rigid transformation parameters by minimizing joint center error. These scale factors were not applied to the data used in downstream analyses but instead were recorded as a diagnostic measure of potential scale bias between systems.

A scale factor of  $s = 1.0$  indicates no global scaling difference between systems, whereas deviations from 1.0 reflect a uniform expansion or contraction of the reconstructed markerless skeleton relative to the reference.

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## Author contributions

**Aaron [Surname]:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.  
**Jonathan Matthis:** Conceptualization, Software, Supervision, Writing – review and editing.

## Competing interests

The authors declare that no competing interests exist.

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## Data availability

All data underlying the figures are available at [repository + DOI].

## Code availability

FreeMoCap is open source at <https://github.com/freemocap>.

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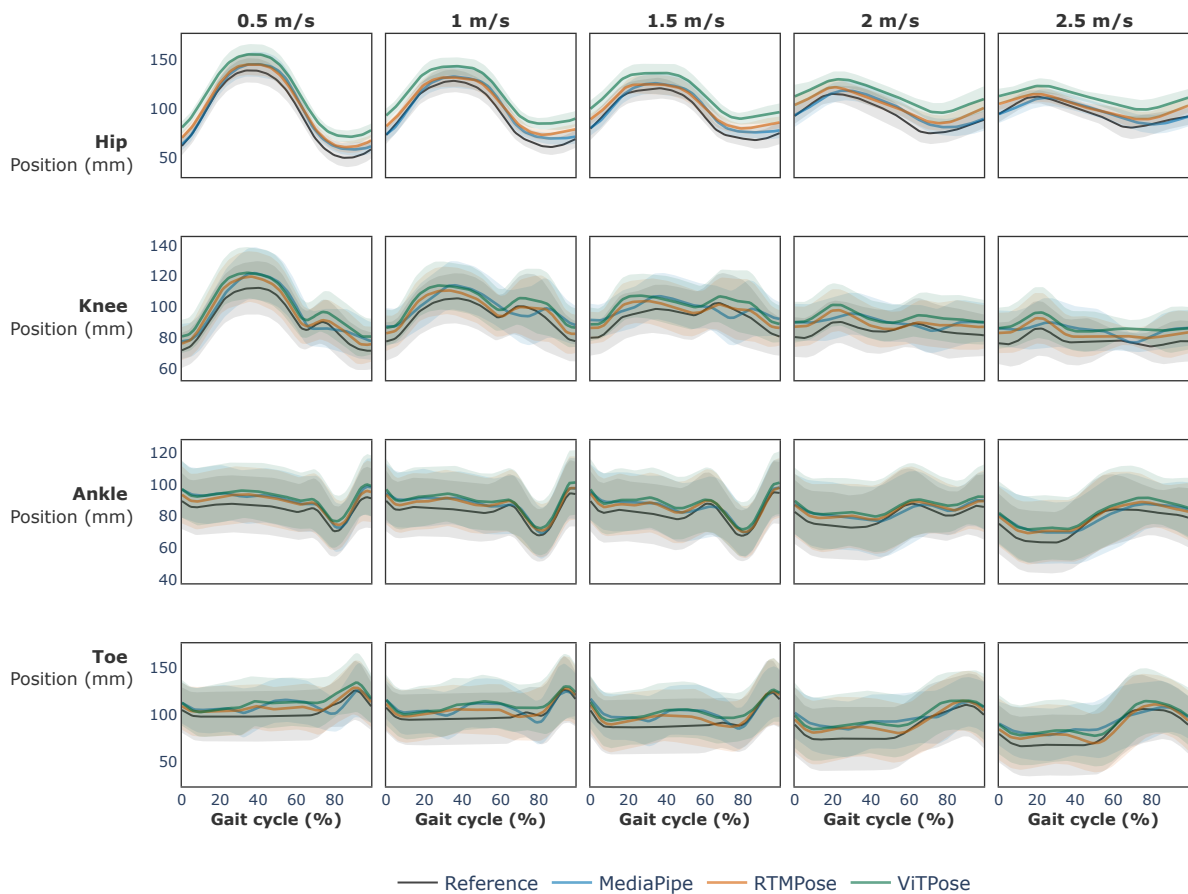
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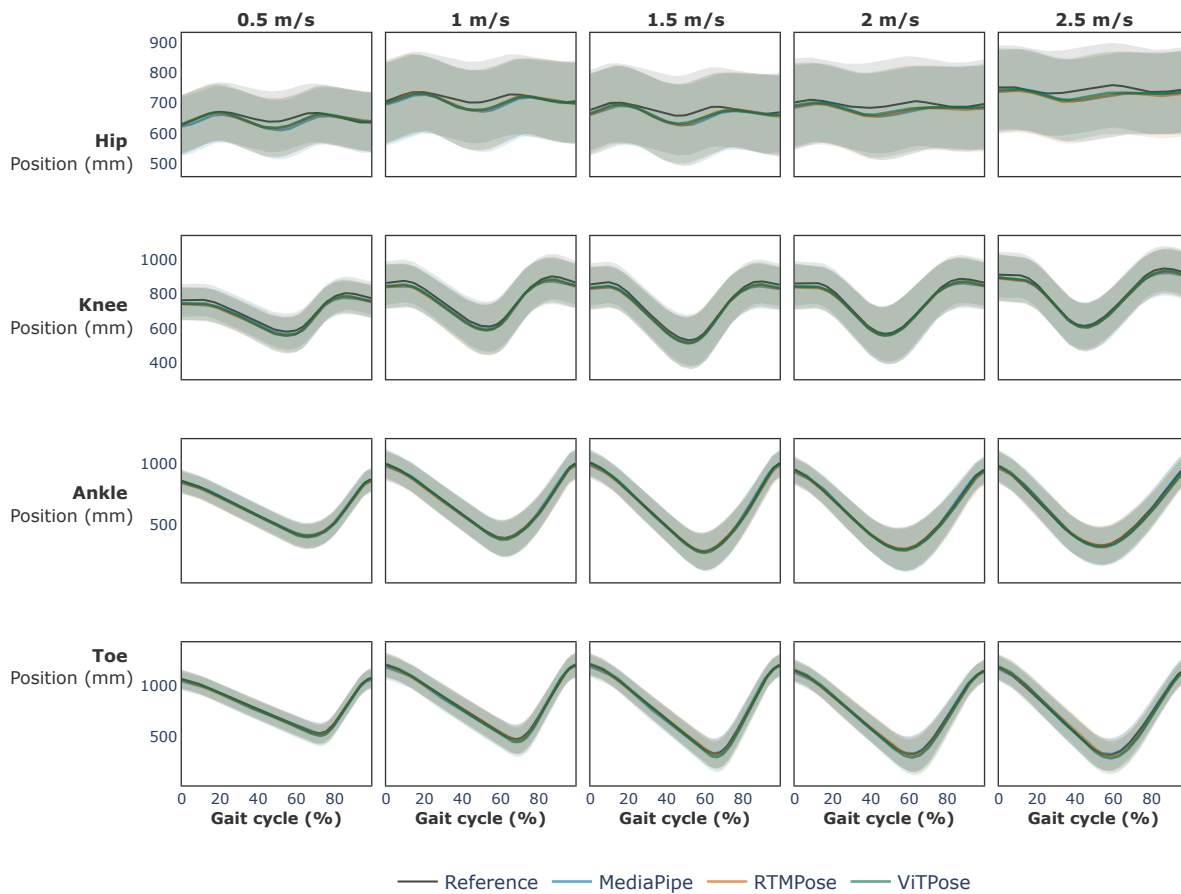
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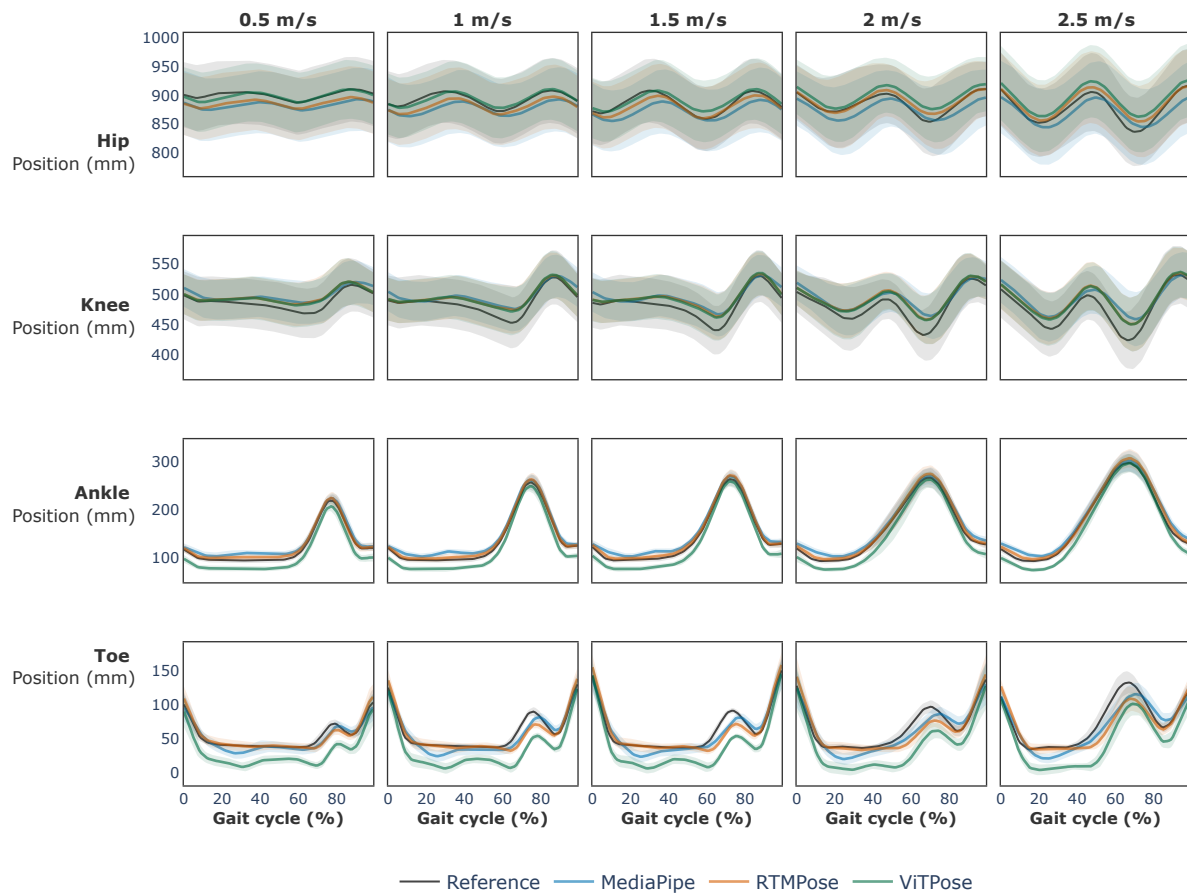
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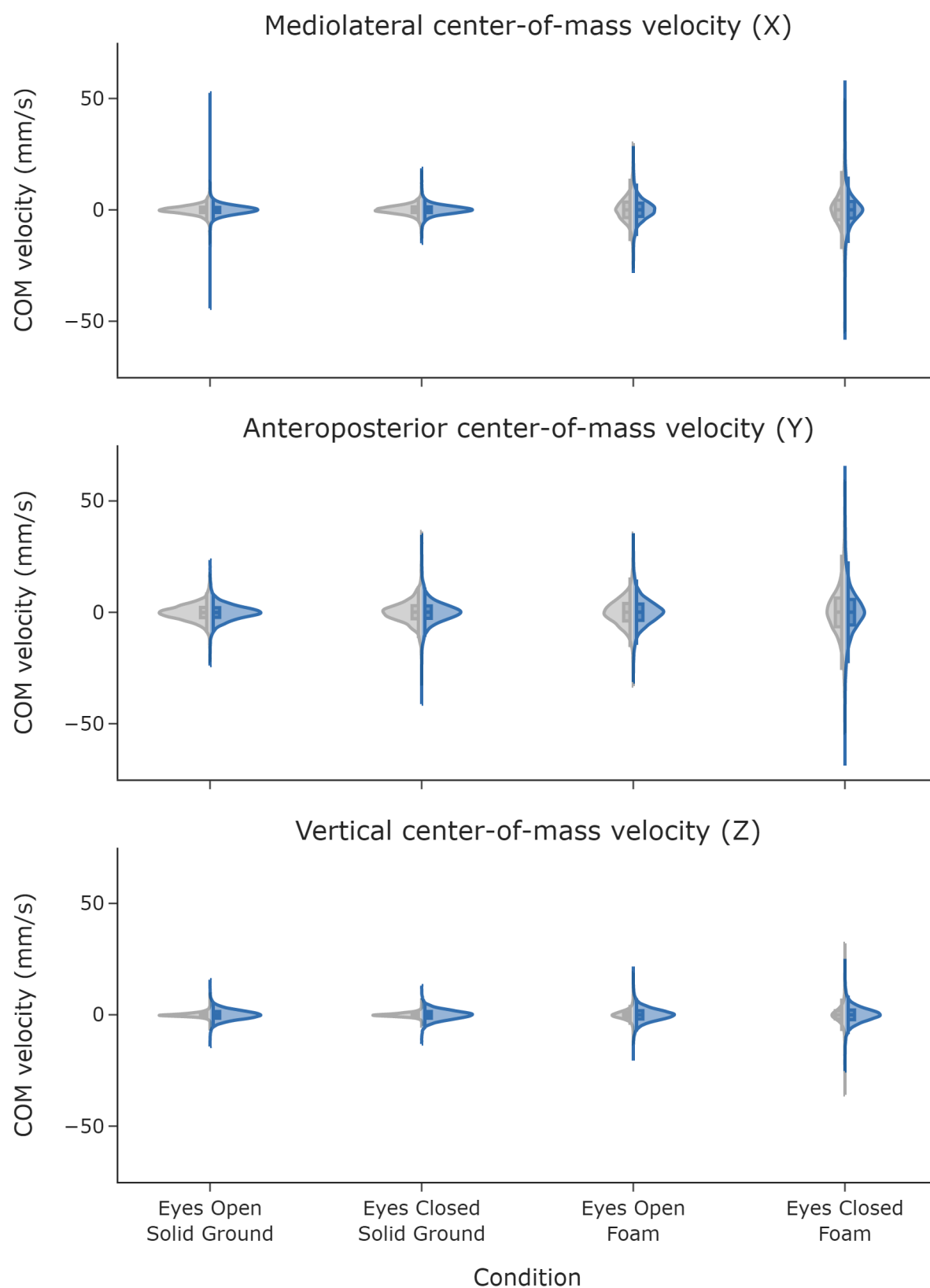
1001 **Figure 4—figure supplement 1.** Comparison of lower-limb (hip, knee, ankle, toe) joint center trajectories in the  
1002 **mediolateral (X) direction** across walking speeds (0.50-2.50 m/s). Trajectories derived from MediaPipe (blue), RTMPose  
1003 (orange), and ViTPose (green) are shown alongside the marker-based reference (grey). Shaded error bars indicate  $\pm 1$  SD  
1004 across strides. The gait cycle is normalized to 0-100%.



**Figure 4—figure supplement 2.** Comparison of lower-limb (hip, knee, ankle, toe) joint center trajectories in the **antero-posterior (Y) direction** across walking speeds (0.50-2.50 m/s). Trajectories derived from MediaPipe (blue), RTMPose (orange), and ViTPose (green) are shown alongside the marker-based reference (grey). Shaded error bars indicate  $\pm 1$  SD across strides. The gait cycle is normalized to 0-100%.



**Figure 4—figure supplement 3.** Comparison of lower-limb (hip, knee, ankle, toe) joint center trajectories in the **vertical (Z) direction** across walking speeds (0.50-2.50 m/s). Trajectories derived from MediaPipe (blue), RTMPose (orange), and ViTPose (green) are shown alongside the marker-based reference (grey). Shaded error bars indicate  $\pm 1$  SD across strides. The gait cycle is normalized to 0-100%.



**Figure 8—figure supplement 1.** Center-of-mass velocity distributions by direction and balance condition, MediaPipe-derived versus the marker-based reference. Distributions match closely in the mediolateral and anteroposterior directions, with a consistently longer tail in the MediaPipe data; the largest discrepancy is in the vertical direction during solid-ground conditions.

## Appendix 1 — Supplementary Gait Data

**Appendix 1—table 1.** Trajectory RMSE per pose estimation backend across speeds — ML (mm). RMSE was calculated per gait-cycle-normalized stride and averaged across strides within each trial. Values represent mean  $\pm$  SD across all participants and trials compared to the marker-based reference.

**Table 3.** Trajectory RMSE per pose estimation backend across speeds — ML (mm). Values represent mean  $\pm$  SD RMSE across all participants and trials compared to the marker-based reference.

| Joint | Backend   | 0.5 m/s         | 1 m/s           | 1.5 m/s         | 2 m/s          | 2.5 m/s        |
|-------|-----------|-----------------|-----------------|-----------------|----------------|----------------|
| Hip   | MediaPipe | 9.6 $\pm$ 7.1   | 9.9 $\pm$ 6.6   | 10.2 $\pm$ 6.1  | 10.2 $\pm$ 5.6 | 8.6 $\pm$ 4.9  |
|       | RTMPose   | 10.4 $\pm$ 8.1  | 10.7 $\pm$ 8.2  | 10.8 $\pm$ 7.8  | 10.8 $\pm$ 7.8 | 9.0 $\pm$ 6.7  |
|       | ViTPose   | 18.9 $\pm$ 10.7 | 19.8 $\pm$ 10.5 | 20.1 $\pm$ 10.4 | 19.2 $\pm$ 9.8 | 16.2 $\pm$ 9.3 |
| Knee  | MediaPipe | 7.0 $\pm$ 2.1   | 7.8 $\pm$ 1.9   | 8.0 $\pm$ 1.7   | 6.9 $\pm$ 1.6  | 7.7 $\pm$ 1.8  |
|       | RTMPose   | 5.5 $\pm$ 2.5   | 5.1 $\pm$ 2.5   | 5.2 $\pm$ 2.4   | 5.7 $\pm$ 2.2  | 5.7 $\pm$ 2.9  |
|       | ViTPose   | 9.1 $\pm$ 2.1   | 8.3 $\pm$ 2.5   | 8.2 $\pm$ 2.6   | 8.9 $\pm$ 2.5  | 9.3 $\pm$ 2.7  |
| Ankle | MediaPipe | 5.9 $\pm$ 1.4   | 5.3 $\pm$ 1.3   | 5.3 $\pm$ 1.1   | 4.9 $\pm$ 1.2  | 5.5 $\pm$ 1.4  |
|       | RTMPose   | 4.8 $\pm$ 1.7   | 4.2 $\pm$ 1.4   | 4.1 $\pm$ 1.4   | 4.4 $\pm$ 1.4  | 4.7 $\pm$ 1.7  |
|       | ViTPose   | 7.3 $\pm$ 1.9   | 6.5 $\pm$ 1.6   | 6.3 $\pm$ 1.6   | 6.5 $\pm$ 1.7  | 6.9 $\pm$ 2.1  |
| Toe   | MediaPipe | 11.8 $\pm$ 2.4  | 11.7 $\pm$ 2.3  | 12.4 $\pm$ 3.5  | 13.6 $\pm$ 4.8 | 12.4 $\pm$ 4.9 |
|       | RTMPose   | 7.7 $\pm$ 2.7   | 7.6 $\pm$ 1.7   | 8.0 $\pm$ 1.6   | 8.6 $\pm$ 2.0  | 8.1 $\pm$ 2.3  |
|       | ViTPose   | 11.5 $\pm$ 3.2  | 11.6 $\pm$ 3.1  | 11.6 $\pm$ 2.6  | 12.1 $\pm$ 2.5 | 10.8 $\pm$ 2.9 |

**Appendix 1—table 2.** Trajectory RMSE per pose estimation backend across speeds — AP (mm). RMSE was calculated per gait-cycle-normalized stride and averaged across strides within each trial. Values represent mean  $\pm$  SD across all participants and trials compared to the marker-based reference.

**Table 4.** Trajectory RMSE per pose estimation backend across speeds — AP (mm). Values represent mean  $\pm$  SD RMSE across all participants and trials compared to the marker-based reference.

| Joint | Backend   | 0.5 m/s        | 1 m/s          | 1.5 m/s        | 2 m/s          | 2.5 m/s         |
|-------|-----------|----------------|----------------|----------------|----------------|-----------------|
| Hip   | MediaPipe | 19.1 $\pm$ 8.2 | 20.3 $\pm$ 8.0 | 22.5 $\pm$ 8.5 | 21.6 $\pm$ 7.0 | 22.0 $\pm$ 6.5  |
|       | RTMPose   | 18.4 $\pm$ 8.4 | 19.2 $\pm$ 8.9 | 21.4 $\pm$ 9.4 | 21.5 $\pm$ 8.8 | 22.5 $\pm$ 6.8  |
|       | ViTPose   | 16.8 $\pm$ 8.7 | 18.8 $\pm$ 7.7 | 20.1 $\pm$ 7.9 | 19.5 $\pm$ 7.4 | 19.6 $\pm$ 7.0  |
| Knee  | MediaPipe | 19.0 $\pm$ 4.9 | 19.8 $\pm$ 4.4 | 20.5 $\pm$ 4.6 | 18.8 $\pm$ 6.2 | 18.4 $\pm$ 6.4  |
|       | RTMPose   | 21.9 $\pm$ 4.9 | 21.8 $\pm$ 4.3 | 20.6 $\pm$ 4.5 | 17.7 $\pm$ 5.1 | 16.7 $\pm$ 5.0  |
|       | ViTPose   | 19.3 $\pm$ 3.9 | 18.9 $\pm$ 3.8 | 18.0 $\pm$ 4.5 | 15.4 $\pm$ 5.7 | 14.7 $\pm$ 4.9  |
| Ankle | MediaPipe | 7.9 $\pm$ 1.8  | 10.0 $\pm$ 2.2 | 13.2 $\pm$ 3.5 | 14.3 $\pm$ 4.0 | 16.9 $\pm$ 7.8  |
|       | RTMPose   | 9.8 $\pm$ 2.5  | 10.5 $\pm$ 2.4 | 12.1 $\pm$ 3.6 | 13.5 $\pm$ 7.1 | 16.6 $\pm$ 7.7  |
|       | ViTPose   | 7.9 $\pm$ 2.8  | 10.1 $\pm$ 3.1 | 12.6 $\pm$ 4.4 | 15.4 $\pm$ 6.6 | 18.9 $\pm$ 6.1  |
| Toe   | MediaPipe | 12.9 $\pm$ 2.9 | 14.8 $\pm$ 4.3 | 17.1 $\pm$ 6.0 | 15.2 $\pm$ 6.3 | 14.5 $\pm$ 7.8  |
|       | RTMPose   | 11.0 $\pm$ 2.6 | 10.9 $\pm$ 4.2 | 13.5 $\pm$ 6.8 | 16.5 $\pm$ 9.4 | 23.1 $\pm$ 11.6 |
|       | ViTPose   | 13.6 $\pm$ 2.4 | 17.5 $\pm$ 3.1 | 21.7 $\pm$ 5.8 | 24.7 $\pm$ 8.1 | 29.2 $\pm$ 9.2  |

**Appendix 1—table 3.** Trajectory RMSE per pose estimation backend across speeds — Vertical (mm). RMSE was calculated per gait-cycle-normalized stride and averaged across strides within each trial. Values represent mean  $\pm$  SD across all participants and trials compared to the marker-based reference

**Table 5.** Trajectory RMSE per pose estimation backend across speeds — Vertical (mm). Values represent mean  $\pm$  SD RMSE across all participants and trials compared to the marker-based reference.

| Joint | Backend   | 0.5 m/s         | 1 m/s           | 1.5 m/s        | 2 m/s           | 2.5 m/s         |
|-------|-----------|-----------------|-----------------|----------------|-----------------|-----------------|
| Hip   | MediaPipe | 21.2 $\pm$ 13.7 | 21.0 $\pm$ 11.3 | 20.8 $\pm$ 9.6 | 19.5 $\pm$ 8.1  | 19.6 $\pm$ 4.8  |
|       | RTMPose   | 19.7 $\pm$ 9.8  | 18.4 $\pm$ 7.7  | 16.5 $\pm$ 5.7 | 14.6 $\pm$ 8.6  | 13.7 $\pm$ 10.6 |
|       | ViTPose   | 14.9 $\pm$ 9.1  | 14.1 $\pm$ 7.8  | 14.7 $\pm$ 7.9 | 18.3 $\pm$ 11.4 | 19.6 $\pm$ 13.9 |
| Knee  | MediaPipe | 12.6 $\pm$ 7.2  | 14.5 $\pm$ 6.3  | 15.9 $\pm$ 5.3 | 18.6 $\pm$ 5.3  | 21.1 $\pm$ 3.9  |
|       | RTMPose   | 11.3 $\pm$ 5.3  | 11.9 $\pm$ 5.1  | 13.5 $\pm$ 4.7 | 15.2 $\pm$ 4.1  | 16.9 $\pm$ 3.5  |
|       | ViTPose   | 10.0 $\pm$ 4.3  | 10.7 $\pm$ 4.2  | 12.1 $\pm$ 4.0 | 14.1 $\pm$ 3.4  | 16.3 $\pm$ 3.5  |
| Ankle | MediaPipe | 10.6 $\pm$ 4.3  | 10.7 $\pm$ 4.0  | 11.1 $\pm$ 4.3 | 10.4 $\pm$ 4.4  | 11.1 $\pm$ 4.1  |
|       | RTMPose   | 7.3 $\pm$ 3.0   | 6.9 $\pm$ 2.8   | 7.2 $\pm$ 3.0  | 7.0 $\pm$ 3.0   | 7.9 $\pm$ 3.2   |
|       | ViTPose   | 17.8 $\pm$ 5.6  | 17.5 $\pm$ 5.3  | 16.5 $\pm$ 5.2 | 16.1 $\pm$ 4.4  | 15.7 $\pm$ 3.9  |
| Toe   | MediaPipe | 8.3 $\pm$ 2.4   | 11.0 $\pm$ 2.2  | 11.5 $\pm$ 1.8 | 12.5 $\pm$ 1.8  | 13.9 $\pm$ 2.2  |
|       | RTMPose   | 7.0 $\pm$ 2.1   | 10.5 $\pm$ 2.0  | 11.9 $\pm$ 1.9 | 13.4 $\pm$ 2.0  | 16.2 $\pm$ 2.3  |
|       | ViTPose   | 24.9 $\pm$ 3.6  | 27.5 $\pm$ 3.6  | 27.5 $\pm$ 3.1 | 30.8 $\pm$ 2.3  | 32.5 $\pm$ 2.1  |

**Appendix 1—table 4.** Bland-Altman agreement statistics for spatial gait parameters (stride length, step length) stratified by walking speed. Bias and limits of agreement (LoA) are reported in millimeters. ICC = intraclass correlation coefficient (2,1) with 95% confidence interval.

| Metric             | Speed   | Tracker   | Bias  | Lower LoA | Upper LoA | ICC (95% CI)         |
|--------------------|---------|-----------|-------|-----------|-----------|----------------------|
| Stride Length (mm) | 0.5 m/s | MediaPipe | +0.25 | −27.83    | +28.33    | 0.987 (0.980, 0.990) |
|                    |         | RTMPose   | −0.20 | −18.21    | +17.82    | 0.995 (0.990, 1.000) |
|                    |         | ViTPose   | +0.13 | −18.35    | +18.60    | 0.994 (0.990, 1.000) |
|                    | 1.0 m/s | MediaPipe | −0.23 | −37.29    | +36.83    | 0.962 (0.950, 0.970) |
|                    |         | RTMPose   | −0.14 | −38.79    | +38.51    | 0.958 (0.950, 0.970) |
|                    |         | ViTPose   | −0.01 | −36.21    | +36.20    | 0.963 (0.960, 0.970) |
|                    | 1.5 m/s | MediaPipe | +0.33 | −60.62    | +61.28    | 0.959 (0.950, 0.970) |
|                    |         | RTMPose   | +0.37 | −57.81    | +58.55    | 0.963 (0.960, 0.970) |
|                    |         | ViTPose   | −0.04 | −58.11    | +58.04    | 0.963 (0.960, 0.970) |
|                    | 2.0 m/s | MediaPipe | −1.12 | −79.32    | +77.08    | 0.961 (0.950, 0.970) |
|                    |         | RTMPose   | −1.15 | −68.82    | +66.51    | 0.971 (0.970, 0.980) |
|                    |         | ViTPose   | −0.17 | −72.38    | +72.04    | 0.966 (0.960, 0.970) |
|                    | 2.5 m/s | MediaPipe | −0.21 | −94.94    | +94.52    | 0.889 (0.870, 0.910) |
|                    |         | RTMPose   | −0.60 | −88.37    | +87.16    | 0.904 (0.890, 0.920) |
|                    |         | ViTPose   | −0.16 | −87.75    | +87.42    | 0.905 (0.890, 0.920) |
| Step Length (mm)   | 0.5 m/s | MediaPipe | −0.10 | −35.30    | +35.11    | 0.936 (0.920, 0.950) |
|                    |         | RTMPose   | −0.32 | −19.87    | +19.23    | 0.979 (0.970, 0.980) |
|                    |         | ViTPose   | −0.19 | −21.36    | +20.99    | 0.976 (0.970, 0.980) |
|                    | 1.0 m/s | MediaPipe | +1.09 | −36.54    | +38.73    | 0.875 (0.850, 0.900) |
|                    |         | RTMPose   | +0.20 | −37.34    | +37.73    | 0.877 (0.850, 0.900) |
|                    |         | ViTPose   | −0.69 | −37.95    | +36.58    | 0.881 (0.860, 0.900) |
|                    | 1.5 m/s | MediaPipe | +1.86 | −59.60    | +63.33    | 0.849 (0.820, 0.870) |
|                    |         | RTMPose   | +0.75 | −56.52    | +58.03    | 0.870 (0.850, 0.890) |
|                    |         | ViTPose   | −0.04 | −63.39    | +63.31    | 0.846 (0.820, 0.870) |
|                    | 2.0 m/s | MediaPipe | −0.04 | −78.88    | +78.81    | 0.868 (0.850, 0.890) |
|                    |         | RTMPose   | −0.88 | −71.59    | +69.83    | 0.895 (0.880, 0.910) |
|                    |         | ViTPose   | +0.33 | −76.96    | +77.61    | 0.874 (0.850, 0.890) |
|                    | 2.5 m/s | MediaPipe | +0.64 | −94.13    | +95.40    | 0.733 (0.690, 0.770) |
|                    |         | RTMPose   | −1.03 | −84.40    | +82.33    | 0.789 (0.750, 0.820) |
|                    |         | ViTPose   | +1.04 | −85.97    | +88.06    | 0.776 (0.740, 0.810) |

**Appendix 1—table 5.** Bland-Altman agreement statistics for temporal gait parameters (stance duration, swing duration) stratified by walking speed. Bias and limits of agreement (LoA) are reported in milliseconds. ICC = intraclass correlation coefficient (2,1) with 95% confidence interval.

| Metric               | Speed   | Tracker   | Bias   | Lower LoA | Upper LoA | ICC (95% CI)         |
|----------------------|---------|-----------|--------|-----------|-----------|----------------------|
| Stance Duration (ms) | 0.5 m/s | MediaPipe | +11.19 | −126.04   | +148.43   | 0.861 (0.830, 0.890) |
|                      |         | RTMPose   | +3.12  | −32.34    | +38.57    | 0.990 (0.990, 0.990) |
|                      |         | ViTPose   | +3.60  | −32.97    | +40.16    | 0.989 (0.990, 0.990) |
|                      | 1.0 m/s | MediaPipe | +7.52  | −34.95    | +50.00    | 0.904 (0.860, 0.930) |
|                      |         | RTMPose   | +3.86  | −35.69    | +43.41    | 0.921 (0.900, 0.940) |
|                      |         | ViTPose   | −1.57  | −39.96    | +36.81    | 0.929 (0.910, 0.940) |
|                      | 1.5 m/s | MediaPipe | +4.66  | −35.31    | +44.63    | 0.894 (0.870, 0.910) |
|                      |         | RTMPose   | +1.82  | −33.51    | +37.15    | 0.921 (0.910, 0.930) |
|                      |         | ViTPose   | −3.35  | −39.94    | +33.25    | 0.913 (0.890, 0.930) |
|                      | 2.0 m/s | MediaPipe | +1.77  | −37.38    | +40.92    | 0.935 (0.920, 0.940) |
|                      |         | RTMPose   | +1.03  | −32.90    | +34.95    | 0.952 (0.940, 0.960) |
|                      |         | ViTPose   | −2.17  | −36.34    | +32.01    | 0.950 (0.940, 0.960) |
|                      | 2.5 m/s | MediaPipe | +2.66  | −40.46    | +45.79    | 0.650 (0.590, 0.700) |
|                      |         | RTMPose   | +3.43  | −35.42    | +42.28    | 0.698 (0.650, 0.740) |
|                      |         | ViTPose   | +0.77  | −37.24    | +38.78    | 0.724 (0.680, 0.760) |
| Swing Duration (ms)  | 0.5 m/s | MediaPipe | −15.28 | −68.16    | +37.59    | 0.838 (0.680, 0.910) |
|                      |         | RTMPose   | −3.36  | −38.29    | +31.57    | 0.936 (0.920, 0.950) |
|                      |         | ViTPose   | −3.72  | −40.02    | +32.59    | 0.933 (0.910, 0.950) |
|                      | 1.0 m/s | MediaPipe | −7.70  | −47.87    | +32.47    | 0.637 (0.520, 0.720) |
|                      |         | RTMPose   | −3.95  | −38.12    | +30.22    | 0.733 (0.680, 0.780) |
|                      |         | ViTPose   | +1.49  | −34.34    | +37.31    | 0.729 (0.680, 0.770) |
|                      | 1.5 m/s | MediaPipe | −4.66  | −45.78    | +36.47    | 0.426 (0.340, 0.500) |
|                      |         | RTMPose   | −1.67  | −39.08    | +35.74    | 0.527 (0.460, 0.590) |
|                      |         | ViTPose   | +3.13  | −36.36    | +42.62    | 0.481 (0.410, 0.550) |
|                      | 2.0 m/s | MediaPipe | −1.77  | −42.57    | +39.03    | 0.637 (0.590, 0.680) |
|                      |         | RTMPose   | −1.08  | −36.99    | +34.82    | 0.714 (0.670, 0.750) |
|                      |         | ViTPose   | +2.17  | −35.08    | +39.42    | 0.696 (0.650, 0.740) |
|                      | 2.5 m/s | MediaPipe | −2.73  | −46.56    | +41.10    | 0.567 (0.500, 0.630) |
|                      |         | RTMPose   | −3.64  | −41.83    | +34.55    | 0.664 (0.610, 0.710) |
|                      |         | ViTPose   | −0.98  | −39.79    | +37.83    | 0.664 (0.610, 0.710) |

## Appendix 2 — Supplementary Balance Data

**Appendix 2—table 1.** Center-of-mass postural sway metrics (path length, 95% prediction ellipse area, and 2D mean velocity in the horizontal plane) across balance conditions for each tracker. Values are reported as group mean  $\pm$  SD. EO = eyes open, EC = eyes closed.

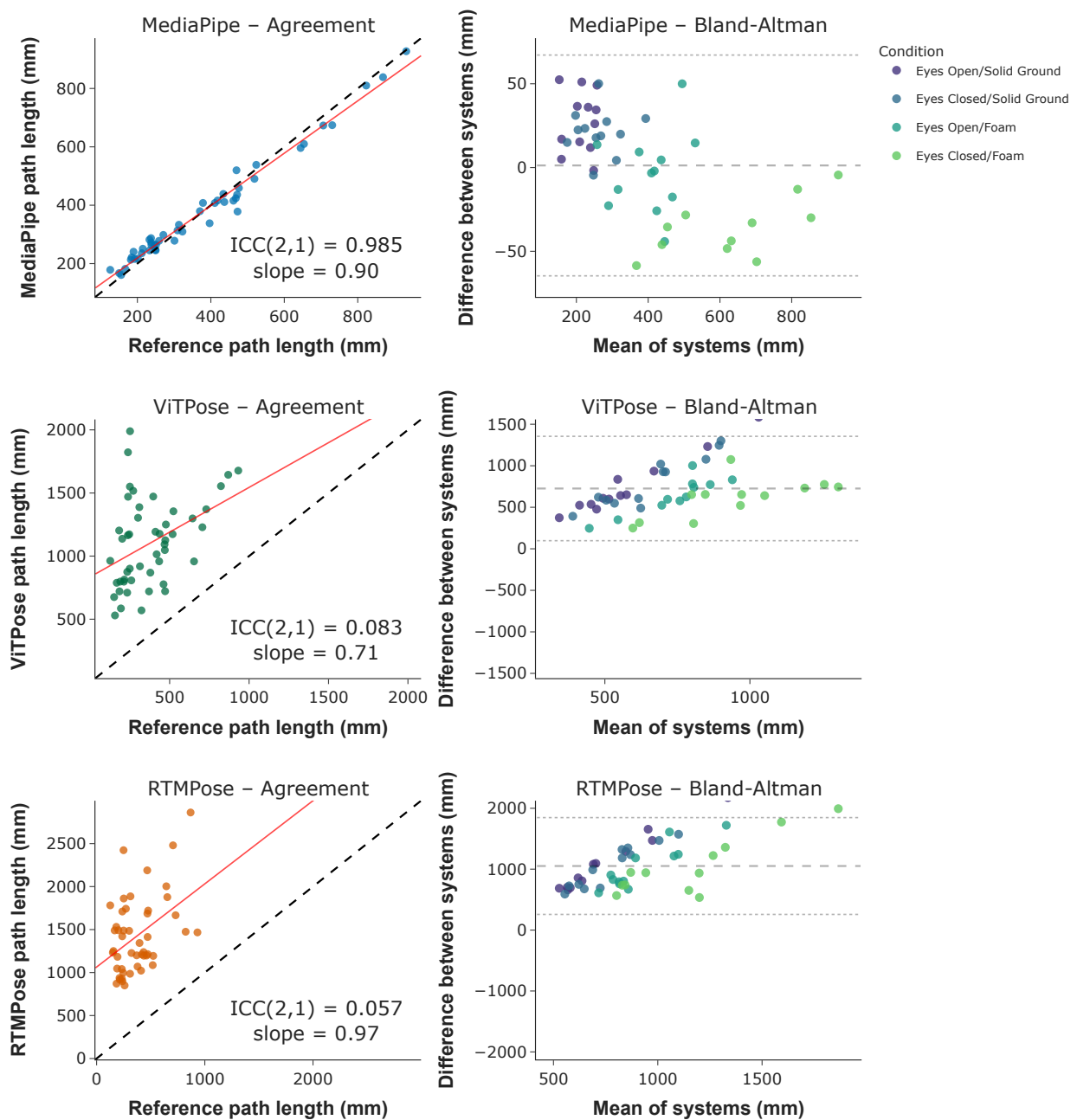
| Condition  | Backend   | Path Length<br>(mm) | Ellipse Area<br>(mm <sup>2</sup> ) | Mean Velocity<br>(mm/s) |
|------------|-----------|---------------------|------------------------------------|-------------------------|
| EO / Solid | Reference | 201.1 $\pm$ 38.7    | 256.8 $\pm$ 111.7                  | 3.63 $\pm$ 0.75         |
|            | MediaPipe | 228.9 $\pm$ 39.3    | 242.4 $\pm$ 111.1                  | 3.54 $\pm$ 0.73         |
|            | RTMPose   | 1299.2 $\pm$ 383.0  | 274.4 $\pm$ 118.5                  | 19.52 $\pm$ 8.09        |
|            | ViTPose   | 951.3 $\pm$ 363.4   | 269.6 $\pm$ 124.7                  | 14.93 $\pm$ 6.80        |
| EC / Solid | Reference | 251.5 $\pm$ 55.3    | 310.5 $\pm$ 150.7                  | 4.61 $\pm$ 1.05         |
|            | MediaPipe | 272.7 $\pm$ 53.2    | 304.6 $\pm$ 160.3                  | 4.45 $\pm$ 1.02         |
|            | RTMPose   | 1298.6 $\pm$ 275.7  | 310.9 $\pm$ 142.2                  | 19.45 $\pm$ 5.41        |
|            | ViTPose   | 1063.5 $\pm$ 277.6  | 331.9 $\pm$ 149.2                  | 17.20 $\pm$ 5.23        |
| EO / Foam  | Reference | 406.9 $\pm$ 81.0    | 971.5 $\pm$ 467.2                  | 7.35 $\pm$ 1.52         |
|            | MediaPipe | 403.8 $\pm$ 87.7    | 938.2 $\pm$ 476.2                  | 6.84 $\pm$ 1.48         |
|            | RTMPose   | 1435.8 $\pm$ 230.7  | 977.7 $\pm$ 481.2                  | 21.77 $\pm$ 4.92        |
|            | ViTPose   | 1139.3 $\pm$ 319.2  | 971.1 $\pm$ 447.7                  | 18.53 $\pm$ 5.79        |
| EC / Foam  | Reference | 639.9 $\pm$ 171.6   | 870.2 $\pm$ 387.3                  | 11.48 $\pm$ 3.22        |
|            | MediaPipe | 599.0 $\pm$ 188.1   | 843.7 $\pm$ 390.5                  | 10.53 $\pm$ 3.37        |
|            | RTMPose   | 1673.8 $\pm$ 428.3  | 897.2 $\pm$ 380.2                  | 26.14 $\pm$ 7.46        |
|            | ViTPose   | 1250.1 $\pm$ 300.7  | 858.7 $\pm$ 373.0                  | 20.54 $\pm$ 5.30        |

**Appendix 2—table 2.** Summary of COM path length agreement metrics across FreeMoCap pose estimation backends compared to Qualisys.

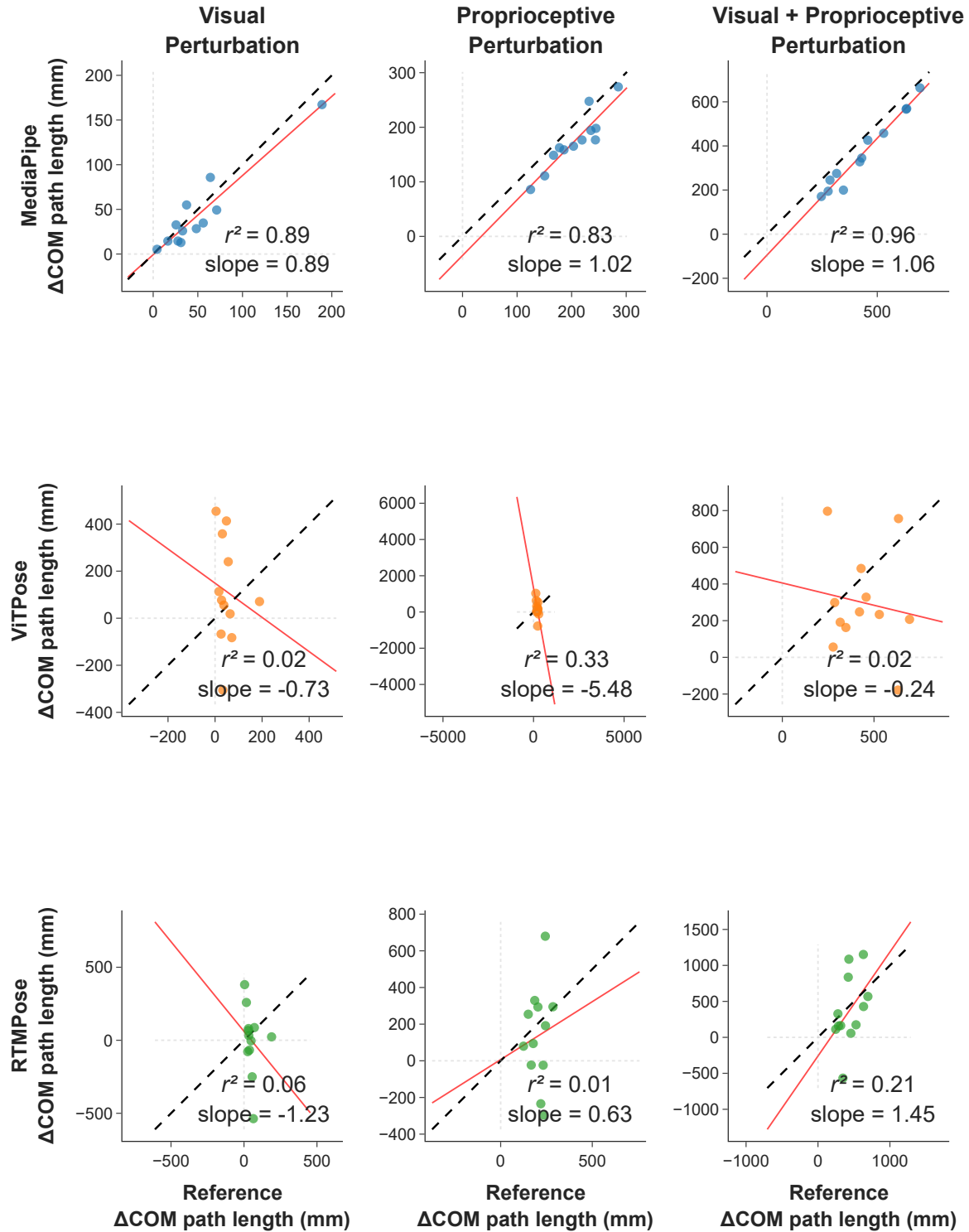
| System    | ICC(2,1) (95% CI)     | Bias (mm) | LoA (mm)          | Slope |
|-----------|-----------------------|-----------|-------------------|-------|
| MediaPipe | 0.985 (0.970, 0.990)  | 1.25      | (-64.63, 67.13)   | 0.90  |
| ViTPose   | 0.083 (-0.060, 0.290) | 726.19    | (97.82, 1354.56)  | 0.71  |
| RTMPose   | 0.057 (-0.040, 0.220) | 1052.02   | (257.06, 1846.98) | 0.97  |

**Appendix 2—table 3.** Summary of COM path length sensitivity metrics across FreeMoCap pose estimation backends. Each cell reports the regression slope and coefficient of determination ( $r^2$ ) for the condition contrast relative to Qualisys.

| System    | Visual<br>Perturbation      | Proprioceptive<br>Perturbation | Visual + Proprioceptive<br>Perturbation |
|-----------|-----------------------------|--------------------------------|-----------------------------------------|
| MediaPipe | slope = 0.89, $r^2$ = 0.89  | slope = 1.02, $r^2$ = 0.83     | slope = 1.06, $r^2$ = 0.96              |
| ViTPose   | slope = -0.73, $r^2$ = 0.02 | slope = -5.48, $r^2$ = 0.33    | slope = -0.24, $r^2$ = 0.02             |
| RTMPose   | slope = -1.23, $r^2$ = 0.06 | slope = 0.63, $r^2$ = 0.01     | slope = 1.45, $r^2$ = 0.21              |



**Appendix 2—figure 1.** Trial-level agreement between markerless and marker-based reference center of mass path length ( $n = 12$ ) per pose estimation backend (Top: Mediapipe; Middle: ViTPose; Bottom: RTMPose). Left: Identity plot with identity line (dashed) and line of best fit (red). Right: Bland-Altman plot of differences between tracker-derived and reference estimates plotted against the mean of the two measurements. Dashed lines represent bias, and dotted lines represent 95% limits of agreement. Colors indicate balance condition.



**Appendix 2—figure 2.** Sensitivity of markerless system center of mass path length to condition-dependent perturbations across all pose estimation backends (Top: MediaPipe; Middle: ViTPose; Bottom: RTMPose). Identity plots are shown with identity line (dashed) and line of best fit (red). Each panel shows trial-level path length differences ( $n = 12$ ) in different conditions, representing system perturbations. Left: Visual perturbation = Eyes Closed (Solid Ground) - Eyes Open (Solid Ground); Middle: Proprioceptive perturbation = Eyes Open (Foam Pad) - Eyes Open (Solid Ground); Right: Visual + Proprioceptive Perturbation = Eyes Closed (Foam Pad) - Eyes Open (Solid Ground).

1187 )

## 1188 **Appendix 3 — Pose estimation backend configurations**

1189 All three pose estimation backends were executed within FreeMoCap through the SkellyTracker inter-  
1190 face. The specific settings for each pose estimation in the SkellyTracker interface are detailed below.

### 1191 **ViTPose.**

1192 Implemented via the `easy_ViTPose` repository on GitHub - with a wrapper implemented in  
1193 SkellyTracker to handle download of necessary models for ViTPose and YOLOv8 (as the person detector)  
1194 from HuggingFace.

```
1195 HF_VIT_REPO = "JunkyByte/easy_ViTPose"  
1196 HF_YOLO_REPO = "ultraalytics/YOLOv8"
```

1197

```
1198 confidence_threshold: float = 0.5  
1199 vit_model: str = "huge" # -> torch/wholebody/vitpose-h-wholebody.pth  
1200 yolo_model: str = "medium" # -> yolov8m.pt  
1201 yolo_size: int = 640  
1202 yolo_step: int = 1  
1203 device: str|None = None
```

### 1204 **RTMPose.**

1205 Implemented via `rtmlib` v0.0.14, using the Wholebody model (COCO-WholeBody 133).

```
1206 # rtmlib v0.0.14 - Wholebody (COCO-WholeBody 133)  
1207 class RTMPoseDetectorConfig(BaseDetectorConfig):  
1208     mode: str = "performance"  
1209     backend: str = "onnxruntime"  
1210     device: str = "cuda"
```

### 1211 **MediaPipe.**

1212 Using MediaPipe Holistic from v0.10.14. This is the legacy Holistic solution, deprecated by Google in  
1213 favor of the Tasks API; we remained on the legacy version because Holistic's integrated single-pass body  
1214 + face + hand topology is the basis of the FreeMoCap processing pipeline and is what these data were  
1215 processed with.

```
1216 # MediaPipe Holistic v0.10.14 (legacy solution)  
1217 MEDIAPIPE_TRACKER_POSTHOC_PRESET = MediapipeDetectorConfig(  
1218     model_complexity=MediapipeModelComplexity.HEAVY, # = 2  
1219     min_detection_confidence=0.5,  
1220     min_tracking_confidence=0.5,  
1221     static_image_mode=False,  
1222     smooth_landmarks=True,  
1223     enable_segmentation=True,  
1224     smooth_segmentation=True,  
1225     refine_face_landmarks=True,  
1226 )
```

1227 One main setting warrants emphasis. The MediaPipe configuration is the only one applying temporal  
1228 smoothing: with `smooth_landmarks = True`, MediaPipe Holistic applies a One Euro filter to the output  
1229 landmarks, whereas RTMPose and ViTPose were run per-frame with no temporal filtering. The One  
1230 Euro parameters are internal to the MediaPipe graph; setting either `static_image_mode = True` or  
1231 `smooth_landmarks = False` disables the filter. These configurations are summarized in the table below.

1232 **Appendix 3—table 1.** Pose-estimation backend configurations. Detectors, keypoint formats, temporal processing; these  
1233 differences bear on the agreement and signal-quality results reported in the main text.

| 1234 |                         | <b>MediaPipe</b>        | <b>RTMPose</b>     | <b>ViTPose</b>      |
|------|-------------------------|-------------------------|--------------------|---------------------|
| 1235 | Library / version       | MediaPipe Holistic      | rtmlib v0.0.14     | easy_ViTPose (Hug-  |
| 1236 |                         | v0.10.14 (legacy)       |                    | gingFace)           |
| 1237 | Person detector         | Internal (MediaPipe)    | RTMDet (bundled in | YOLOv8m             |
| 1238 |                         |                         | rtmlib)            | (yolov8m.pt)        |
| 1239 | Pose model / checkpoint | Holistic, complexity    | Wholebody, mode =  | ViTPose-H, vitpose- |
| 1240 |                         | HEAVY (= 2)             | "performance"      | h-wholebody.pth     |
| 1241 | Native keypoint set     | 33 pose + 468 face + 21 | COCO-WholeBody     | COCO-WholeBody      |
| 1242 |                         | ×2 hand                 | (133)              | (133)               |
| 1243 | Temporal processing     | One Euro filter (via    | none (per-frame)   | none (per-frame)    |
| 1244 |                         | smooth_landmarks)       |                    |                     |
| 1245 | Compute backend         | TensorFlow Lite (inter- | ONNX Runtime       | PyTorch             |
| 1246 |                         | nal)                    |                    |                     |